ANIS Child Application

Technical Documentation

Advanced Network Intelligence & Safety

ANIS Child Application - Android Parental Control System

Version 1.0 - 2026

Kotlin / Jetpack Compose / ONNX Runtime

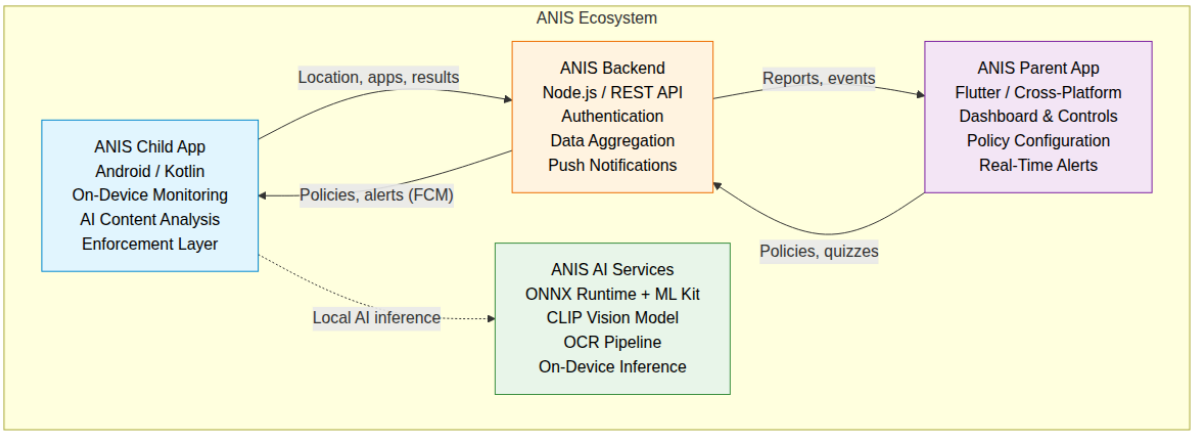
Table of Contents

System Architecture and Design

1. Introduction

1.1 Overview

The ANIS (Advanced Network Intelligence & Safety) Child Application is the device-side enforcement component of the ANIS parental control ecosystem. It is installed on the child's Android device and serves as the protection and monitoring layer of the platform. The application operates in conjunction with three other major components:



- ANIS Parent Application (Flutter): A cross-platform mobile application that provides parents with a dashboard for monitoring their children's digital activity, setting policies, and receiving alerts.
- ANIS Backend Services (Node.js): A central coordination service that mediates communication between parent and child devices, stores policies and usage data, and manages device pairing.
- ANIS AI Services: On-device AI inference engines that perform real-time content moderation using ONNX Runtime and Google ML Kit, operating entirely on the child device to preserve privacy.

Figure 1.1: ANIS ecosystem component interaction diagram.

The child application functions as the enforcement and protection layer. It captures screen content at configurable intervals, analyzes it through a multi-stage AI pipeline, applies protective overlays when inappropriate content is detected, tracks device usage and location, and synchronizes data with the backend for parent visibility. All content analysis is performed on-device, ensuring that sensitive data never leaves the child's device.

1.2 Purpose

The ANIS Child Application is designed to fulfill five primary objectives:

- Protect children from harmful digital content: Real-time screen monitoring using on-device AI classifies content across threat categories (adult, violence, profanity, drugs, weapons, hate, gambling) and applies protective measures when violations are detected.
- Enforce parental control policies: Restrictions configured by parents through the parent application are synchronized to the child device and enforced locally, even when the device is offline.
- Promote healthy digital habits: Screen time limits and scheduled access periods encourage balanced device usage.
- Encourage educational engagement: A reward system motivates children to engage with educational content and earn additional usage time through positive behavior.
- Balance child safety with privacy: Raw screen content (captured images, extracted text) remains on-device and is never transmitted. Analysis results (decisions, threat scores, timestamps) are sent to the backend to provide parents with visibility into their child's digital behavior while preserving privacy.

1.3 Scope

The child application is responsible for all monitoring, analysis, and enforcement functions that execute on the child's device. These responsibilities are divided into local and backend-dependent operations:

Local Operations (performed entirely on-device):

- Screen content capture via MediaProjection API
- OCR text detection via Google ML Kit
- ONNX Runtime model inference for image classification
- Blur overlay display for blocked content
- Accessibility Service-based usage tracking
- PIN validation for offline Settings access
- Background service lifecycle management
- Offline policy caching and local enforcement
- Raw image and text data storage (never transmitted)

Backend-Dependent Operations (require network connectivity):

- Device pairing via QR code
- Location telemetry upload
- Installed applications reporting
- Session and analysis results transmission (decisions, threat scores, timestamps, device stats)
- FCM token registration for push notifications
- Child profile synchronization
- Reward state synchronization
- Policy updates from parent configuration

1.4 Non-Functional Requirements

ID	Requirement	Target
NFR-01	AI analysis latency per frame	< 1000 ms
NFR-02	Memory footprint (sustained)	< 250 MB PSS
NFR-03	Battery drain rate	< 20% per hour
NFR-04	Offline operation capability	24+ hours cached policies
NFR-05	Monitoring service uptime	99.9% (excluding device reboot)
NFR-06	Secure token storage	AES-256-GCM encrypted
NFR-07	API communication	TLS 1.3
NFR-08	Banned word detection accuracy	> 95%
NFR-09	Threat classification false positive rate	< 5%
NFR-10	Data retention	Auto-cleanup after 7 days

2. Technology Selection and Justification

2.1 Development Technology Stack

The ANIS Child Application is built on the Android platform using modern Kotlin-based development practices. The following table summarizes the technologies and libraries selected for each application layer:

Layer	Technology	Purpose
Language	Kotlin 2.0.21	Primary development language with null safety, co
UI Framework	Jetpack Compose + Material 3	Declarative UI with Material You theming and dyna

Architecture	MVVM + Repository Pattern	Separation of concerns with lifecycle-aware compo
Dependency Injection	Hilt 2.51.1	Compile-time dependency injection with applicatio
AI Inference	ONNX Runtime Android 1.25.0	On-device CLIP-based vision model inference with F
OCR	Google ML Kit Text Recognition 16.0.0	On-device text extraction from screen captures
Screen Capture	MediaProjection API + ImageReader	System-level screen capture with VirtualDisplay
Background Work	WorkManager 2.9.1	Deferred and periodic background tasks with const
Local Database	Room 2.6.1	Type-safe SQLite persistence with Flow-based reac
Preferences	EncryptedSharedPreferences 1.1.0	AES-256-GCM encrypted storage for authentication
Networking	Retrofit 2.9.0 + OkHttp 4.12.0	Type-safe HTTP client with interceptor-based authe
Serialization	Kotlinx.serialization 1.6.2	Compile-time JSON serialization for API communica
Push Notifications	Firebase Cloud Messaging 33.12.0	Remote push notification delivery and silent data s
QR Scanning	CameraX 1.4.0 + ML Kit Barcode 17.3.0	Camera-based QR code scanning for device pairing
Location	Google Play Services Location 21.3.0	Fused location provider for periodic GPS telemetry
Export	FastExcel 0.18.1	XLSX workbook generation for session data export
Build System	Gradle 8.13 + AGP 8.13.2 + Kotlin Compose	Modern build configuration with version catalog
CI/CD	GitHub Actions	Automated build, signing, and release publishing

Platform Configuration:

Property	Value
Application ID	`com.anis.child`
Minimum SDK	API 24 (Android 7.0)
Target SDK	API 36 (Android 16)
Compile SDK	API 36
Kotlin JVM Target	11
Gradle Version Catalog	`gradle/libs.versions.toml`

2.2 Why Native Android Instead of Flutter

The decision to implement the ANIS Child Application as a native Android application rather than a cross-platform Flutter alternative was informed by empirical performance data collected from a controlled experiment comparing both implementations of the ANIS AI content moderation pipeline. The experiment, documented in the ANIS research paper [1], tested both implementations under identical conditions (15.5-minute sessions, approximately 900 frames each, same ONNX model and ML Kit pipeline) and provided the following evidence.

Deep Android System Integration

ANIS requires direct interaction with Android system services that are inaccessible or restricted in cross-platform frameworks:

- Accessibility Service: Critical for monitoring app usage, detecting foreground app changes, and intercepting navigation for restriction enforcement. Requires direct Android Service registration and lifecycle management.
- MediaProjection API: Screen capture for content analysis requires MediaProjectionManager and VirtualDisplay, both platform-specific APIs that would require custom plugin development in Flutter.
- UsageStatsManager: Application usage tracking requires system-level permission and direct querying of system usage statistics.
- KeyStore / EncryptedSharedPreferences: Security-sensitive token storage relies on Android hardware-backed keystore, accessible only through Android-specific crypto APIs.

- System Overlays: The blur overlay for blocked content uses TYPE_APPLICATION_OVERLAY window type, requiring direct WindowManager interaction.

In a Flutter implementation, all of the above would require custom native plugins communicating through MethodChannel, introducing serialization overhead of approximately 114 microseconds per call compared to 0.5 microseconds for direct native invocations [2], a difference of over two orders of magnitude.

Performance Summary

A controlled experiment [1] compared both implementations under identical conditions (15.5-minute sessions, Google Pixel 6, ~900 frames each, same ONNX FP16 vision model and ML Kit OCR pipeline).

CPU Utilization:

Metric	Native Android	Flutter
Average CPU Usage	19.42%	18.91%
Variance	Lower	Higher (Dart GC events)

Processing Latency (Per-Phase Breakdown):

Phase	Native Android	Flutter	Ratio
OCR (ML Kit)	176.0 ms	122.5 ms	0.70x
Image Preprocessing	97.0 ms	109.0 ms	1.12x
ONNX Inference	372.0 ms	532.5 ms	1.43x
Total Analysis	645.0 ms	764.0 ms	1.18x

The 1.43x ONNX inference slowdown in Flutter is attributed to Dart FFI marshalling overhead. Flutter's OCR phase is faster (0.70x), likely due to a more efficient ML Kit integration path.

Tail Latency:

Metric	Native Android	Flutter
Maximum frame latency	1047 ms	7829 ms
Frames exceeding 1000 ms	None	12.8%
Frames exceeding 2000 ms	None	0.3% (3 of 999)
95th percentile	~870 ms	1169 ms
99th percentile	~950 ms	1276 ms

Flutter's worst outlier (7829 ms) was caused by an OCR pipeline stall coinciding with Dart garbage collection, a failure mode that does not affect the native Android runtime.

Memory Consumption:

Metric	Native Android	Flutter	Ratio
Average PSS Memory	~180 MB	955.3 MB	~5.3x
Peak PSS Memory	~200 MB	1,160 MB	~5.8x

Flutter memory ranges from 181.6 MB (idle) to 1.16 GB (under load), driven by its bundled rendering engine (Skia/Impeller), Dart VM overhead, and double-layer architecture.

Battery Consumption:

Metric	Native Android	Flutter
Total Discharge (15.5 min)	197 mAh	192 mAh
App Power Contribution	31.6 mAh	~74 mAh (2.3x)

While total battery drain is similar, the app-level contribution is 2.3x higher in Flutter.

Network Data Transfer:

Metric	Native Android	Flutter
Data Transferred	Baseline	5.8x higher

Security and Stability

Native Android provides direct implementation of security-sensitive features without intermediary abstraction layers. Hardware-backed keystore integration, encrypted shared preferences, and certificate pinning are all accessible through documented Android APIs without relying on third-party plugin maintenance. Reduced abstraction layers result in a smaller attack surface.

Plugin Dependency Analysis

Most ANIS functionality would require custom Flutter plugins:

Feature	Native Approach	Flutter Alternative
Screen Capture	MediaProjection (built-in)	Custom MethodChannel plugin
AI Inference	ONNX Runtime Android SDK	flutter_onnxruntime (third-party)
OCR	ML Kit (built-in)	google_mlkit_text_recognition (third-party)
Blur Overlay	WindowManager (built-in)	Custom platform channel
Accessibility	AccessibilityService (built-in)	Custom plugin
Usage Stats	UsageStatsManager (built-in)	Custom plugin
Secure Storage	EncryptedSharedPreferences	flutter_secure_storage (third-party)

Each third-party Flutter plugin represents a maintenance risk: delayed Android API updates, breaking changes, unpatched vulnerabilities, or plugin abandonment.

Conclusion: Native Android development was selected because ANIS is a system-level parental control application, not a conventional mobile application. The application requires deep OS integration, persistent background execution, predictable latency for real-time content moderation, and memory efficiency for continuous operation on resource-constrained devices. While Flutter offers development velocity advantages for standard applications, the empirical evidence demonstrates that the native Android implementation is superior across memory efficiency, latency predictability, and system integration depth, all critical requirements for a child safety platform.

References:

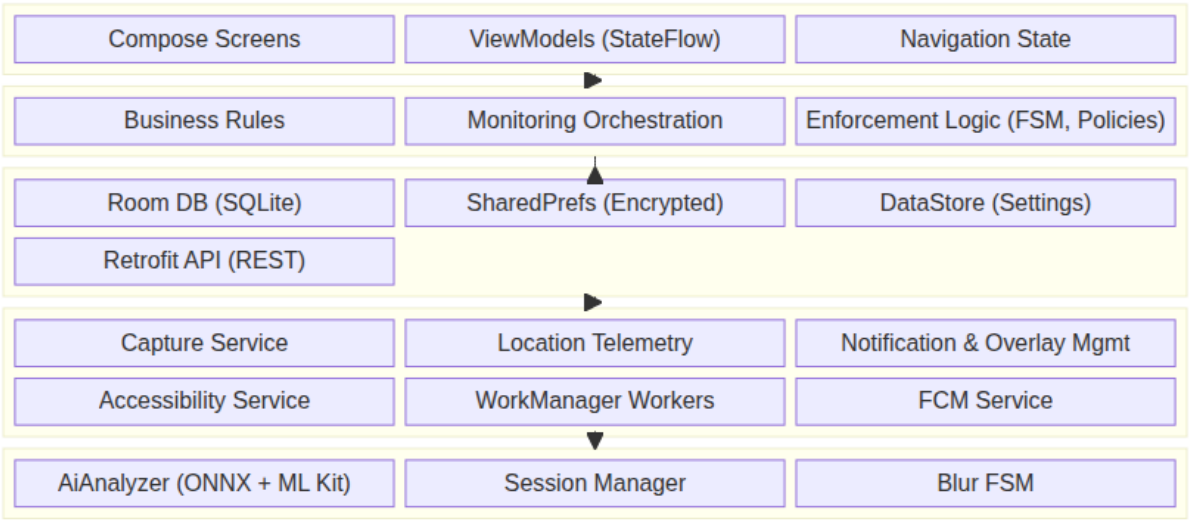
[1] Ahmed Ibrahim et al., "Performance Evaluation of Cross-Platform AI-Powered Content Moderation Systems: Flutter vs Native Android," ANIS Research Paper, 2026.

[2] Pocatilu, P., Vetrici, M., & Despa, M. L., "Performance analysis of cross-platform mobile frameworks for compute-intensive applications," IEEE, 2020.

3. Application Architecture

3.1 Architectural Overview

The ANIS Child Application follows a layered architecture that separates concerns across presentation, domain, data, and service layers. The architecture is designed to support continuous background monitoring while maintaining a responsive user interface and reliable offline operation.



The architecture follows these key principles:

- Unidirectional data flow: UI events flow upward to ViewModels, which invoke repositories and expose state via StateFlow.
- Lifecycle awareness: All components respect Android lifecycle. Services run independently of UI lifecycle.
- Dependency inversion: High-level modules (domain) do not depend on low-level modules (data). Both depend on abstractions.
- Offline-first: Local database is the single source of truth. Network operations synchronize data bidirectionally.

3.2 Architectural Pattern

The application implements the MVVM (Model-View-ViewModel) architectural pattern:

Layer	Component	Responsibility
View	Jetpack Compose Screens	Render UI state, collect user input, observe StateFl
ViewModel	AndroidX ViewModel subclasses	Expose UI state as StateFlow, process user actions,
Model	Data classes, Room entities	Represent application data and business objects

Navigation is managed through Compose state rather than a navigation component. A boolean isLoggedIn state variable determines whether the user sees the PairingScreen or the HomeScreen. Screen-level navigation within the paired state (Home, Settings, PIN gate, Reward) is managed through a sealed class Screen enum.

3.3 Major Layers

Presentation Layer

The presentation layer consists of Jetpack Compose screens that observe ViewModel state and emit user interactions:

- Composable screens: Each screen is a @Composable function that receives state and lambda callbacks. Screens do not hold business logic.
- ViewModels: Each ViewModel extends AndroidViewModel and exposes state as StateFlow. They manage coroutine scopes and lifecycle-aware operations.
- UI state management: State is modeled as sealed classes (PairingUiState, SessionState) ensuring that all possible states are represented.
- Theming: Material 3 theming with dynamic color support on Android 12+, custom light and dark palettes, and consistent typography.

Domain Layer

The domain layer contains business logic, monitoring orchestration, and enforcement rules:

- Business rules: Parental control policies (screen time limits, app restrictions, content filtering rules) define what actions the enforcement layer takes.
- Monitoring orchestration: The TelemetryManager coordinates location collection from FusedLocationProviderClient and delegates to WorkManager for background upload.
- Enforcement logic: The blur overlay FSM (finite state machine) manages transitions between SAFE, PENDING_BLUR, BLURRED, and PENDING_RELEASE states based on consecutive analysis results.
- Session management: The SessionManager orchestrates capture sessions, manages lifecycle states (Idle, Active, PermissionRequired, MediaProjectionRequired, Error), and tracks device resource metrics (battery, CPU, RAM).

Data Layer

The data layer manages all persistent storage and network communication:

- Room Database: Stores location telemetry (location_telemetry table), AI analysis results (analysis_results table), session data (sessions table), and planned entities for reward state.
- EncryptedSharedPreferences: Securely stores authentication tokens (access token, FCM token), child profile data, and PIN hash. Uses AES-256-GCM encryption.
- DataStore: Stores user preferences such as theme selection, monitoring enabled state, and session configuration.
- Repository pattern: Each data source is wrapped in a repository that exposes a clean API to the domain layer. Repositories coordinate between local and remote data sources.
- Retrofit + OkHttp: Type-safe REST client with interceptor-based authentication (AuthInterceptor), request/response logging (AppLoggingInterceptor), and kotlinx.serialization converter.

Service Layer

The service layer manages all background and foreground operations that run independently of the UI:

- Accessibility Service: Monitors system-wide events including app launches, navigation changes, and usage patterns. Enables real-time policy enforcement and usage tracking.
- Capture Service (Foreground Service): Runs in the foreground with a persistent notification. Manages the MediaProjection session, frame capture loop, and AI analysis pipeline.
- Location Telemetry Worker (WorkManager): Periodically (hourly) uploads stored location data with exponential backoff retry.
- FCM Service (FirebaseMessagingService): Handles push notification delivery, FCM token registration and refresh, and triggers immediate location sync on silent push commands.
- Notification Management: Manages foreground service notifications, blocked content alerts, and status indicators.

Core / AI Layer

The AI layer is the computational engine of the application:

- AiAnalyzer: Loads the FP16 quantized CLIP model from assets, performs OCR via ML Kit, runs ONNX inference, and computes embedding similarity for threat classification.
- SessionManager: Configures session parameters (interval, thresholds), manages session lifecycle, tracks device resource metrics.
- Blur FSM: Implements the finite state machine that debounces blur overlay transitions to prevent flickering from transient content.

3.4 Dependency Injection Architecture

Hilt manages the dependency graph across all layers:

- Application component (@HiltAndroidApp): Singleton-scoped dependencies including database, network client, and AI analyzer.

- Activity component (@AndroidEntryPoint): Activity-scoped dependencies and ViewModel providers.
- Service component: Service-scoped dependencies for foreground services.
- ViewModelComponent: ViewModel-scoped dependencies automatically provided by Hilt.

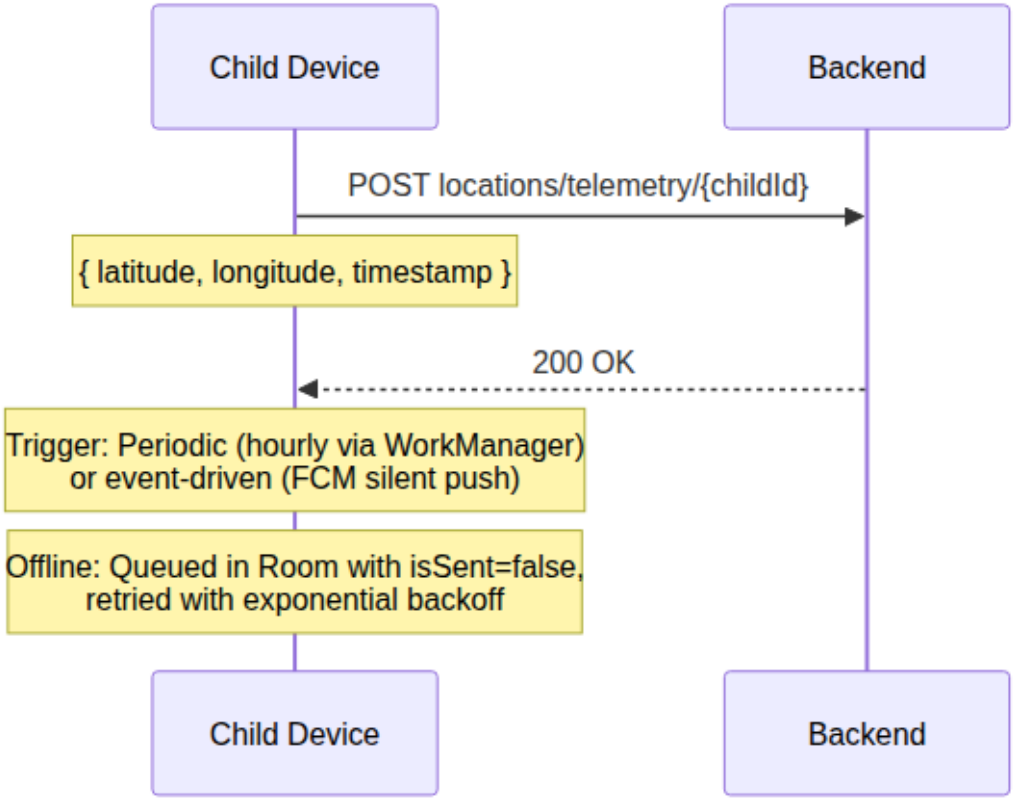
Refer to Section 9 (Dependency Injection with Hilt) for a detailed breakdown of module organization and provided dependencies.

4. Data Flow

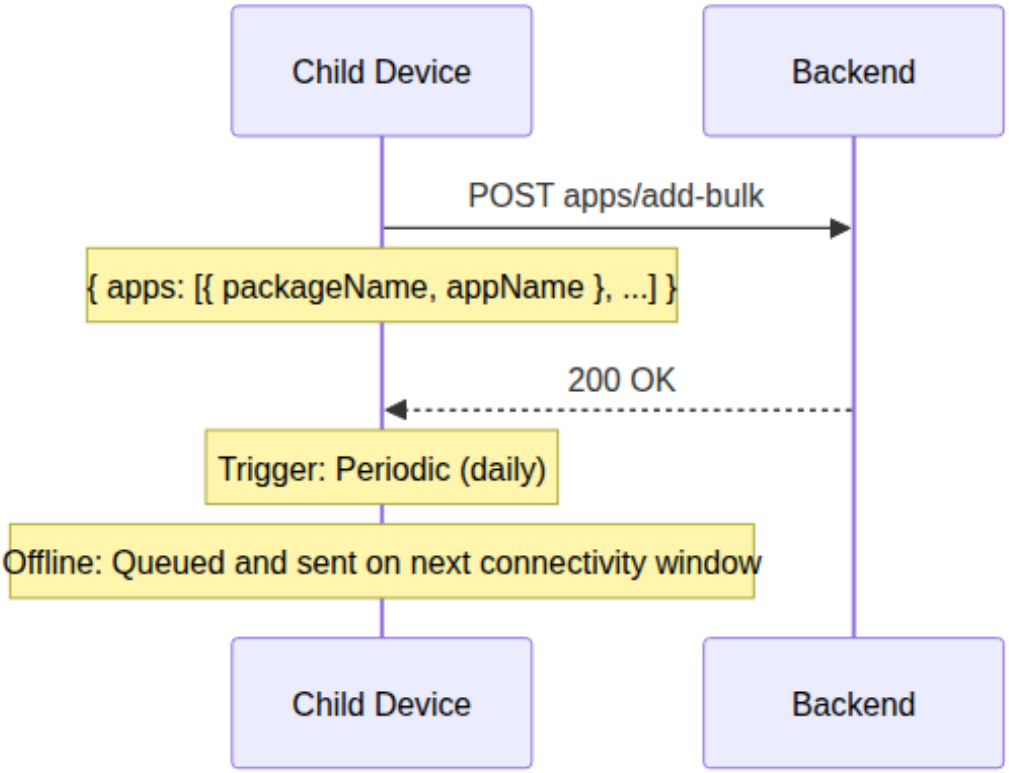
4.1 Device to Backend Flow

The following data types flow from the child device to the backend.

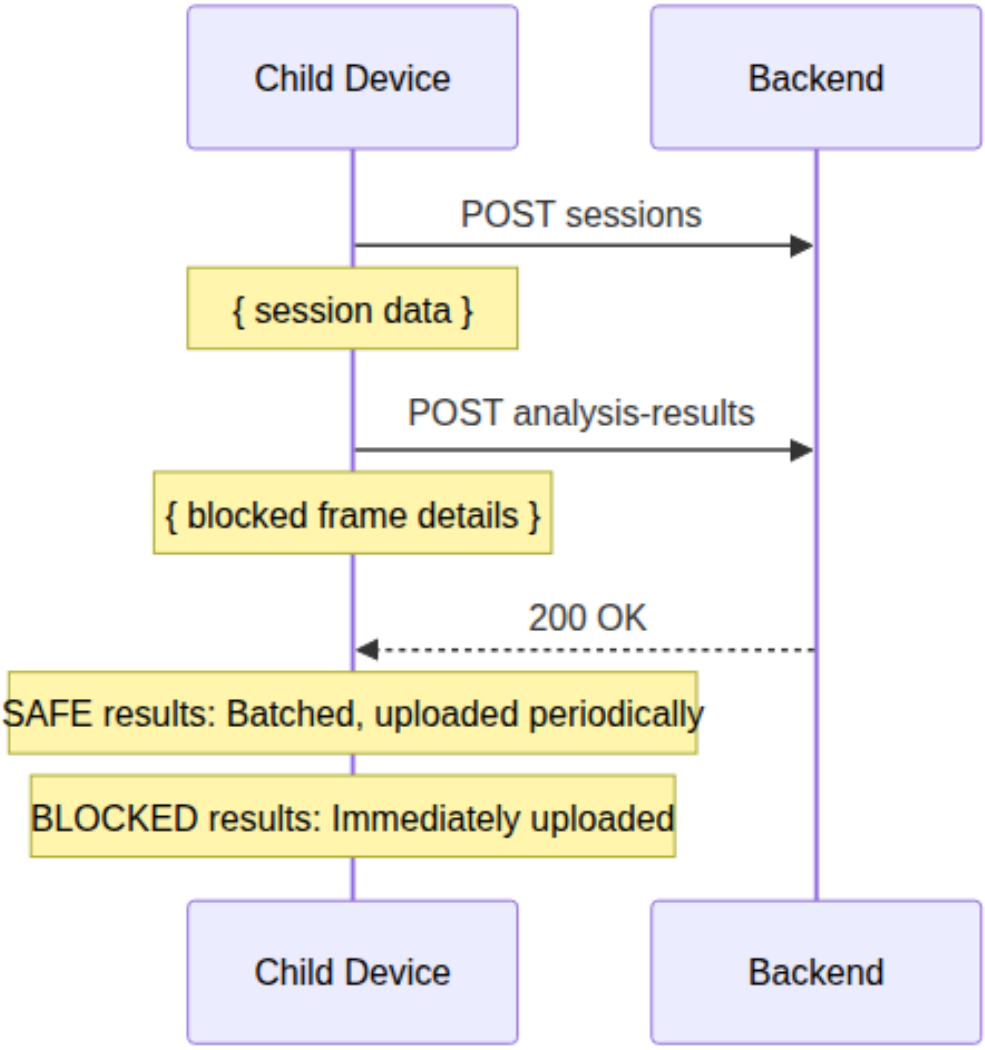
Location Telemetry



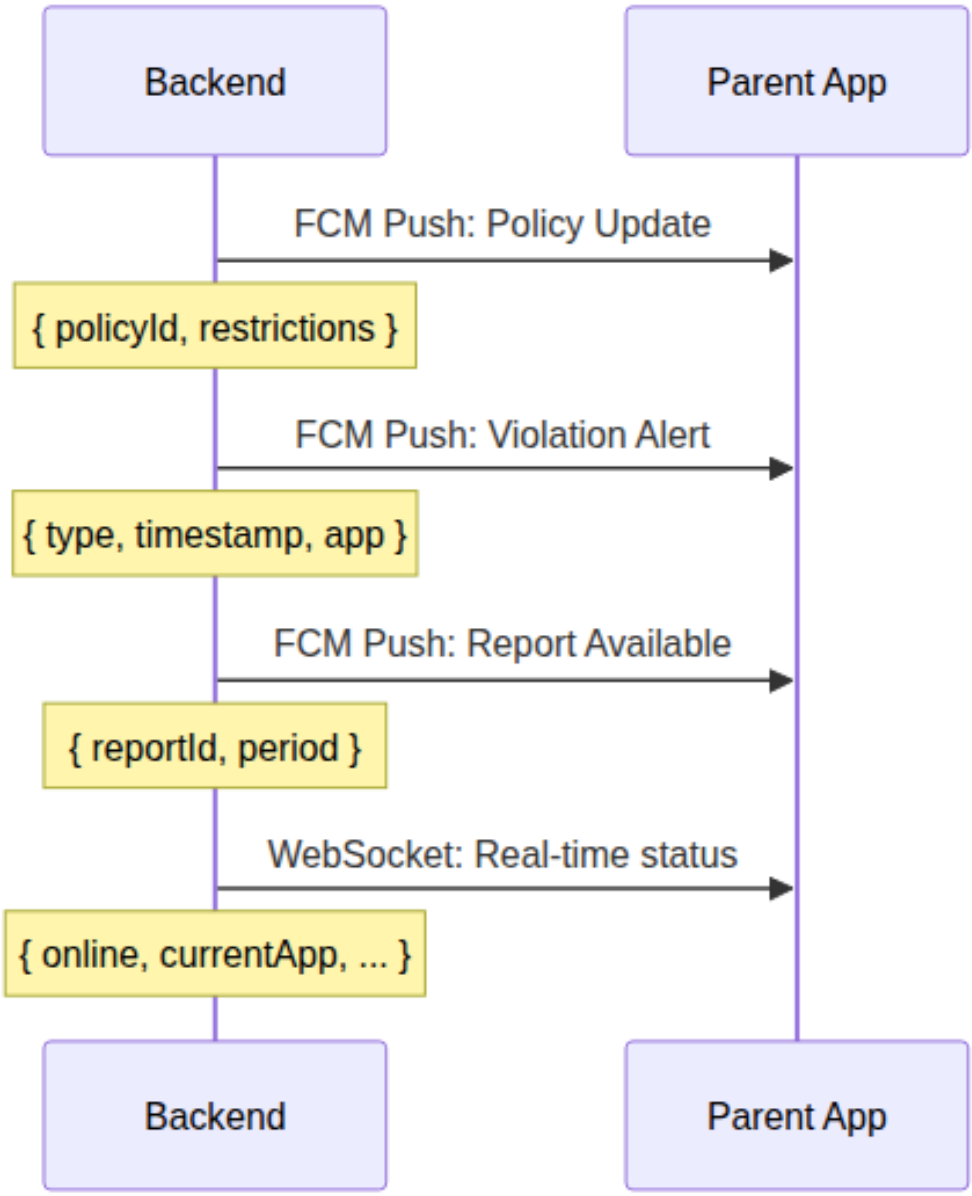
Installed Applications



Analysis Results



4.2 Backend to Parent Application Flow



The parent application receives real-time updates through:

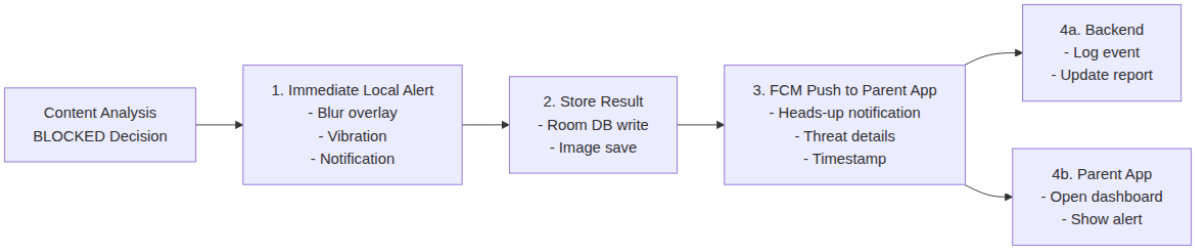
- FCM push notifications for immediate alerts (violations, policy sync).
- REST API polling for periodic data (usage reports, location history).
- Planned WebSocket integration for real-time device status.

4.3 Content Analysis Flow

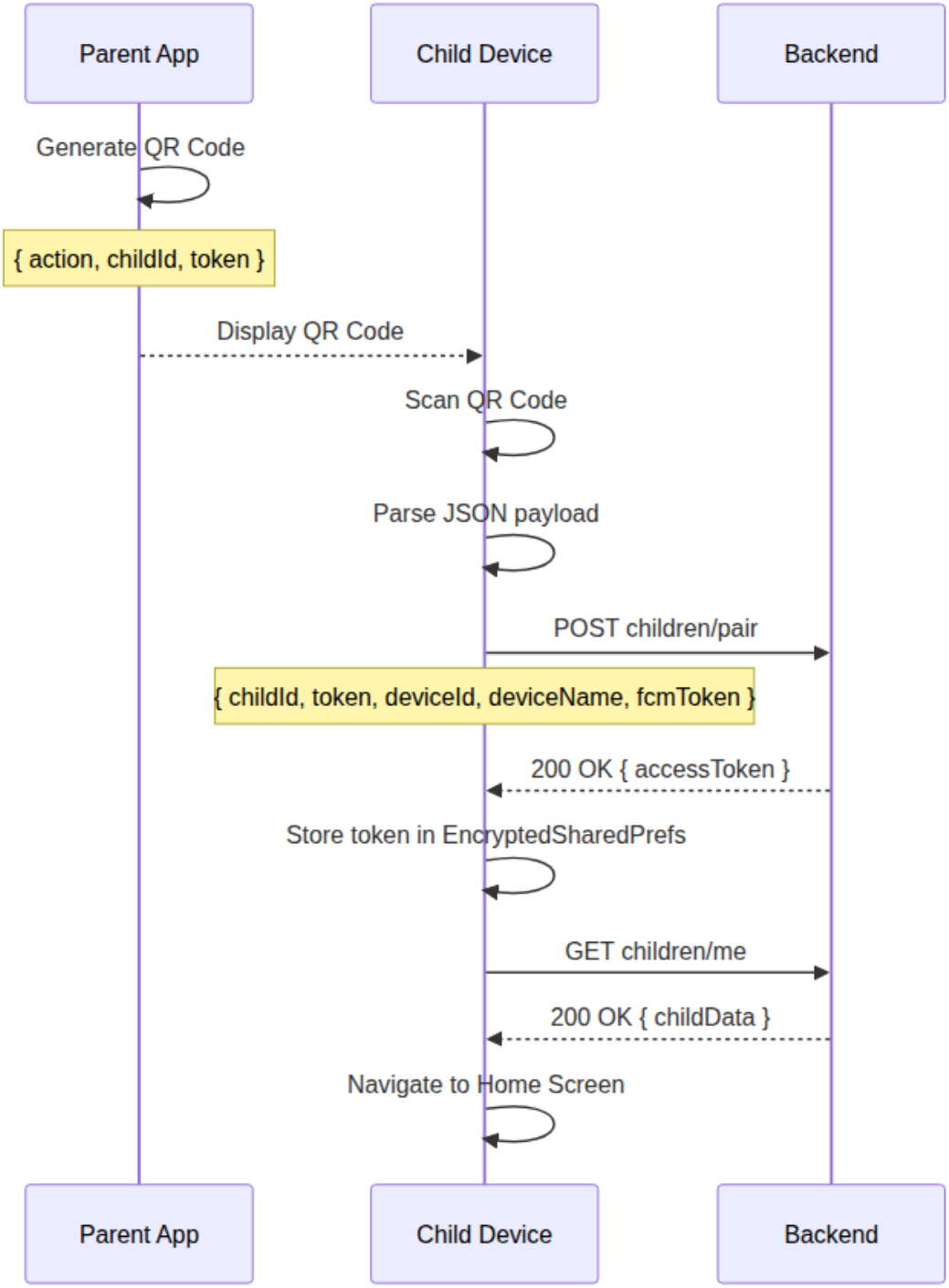
The captured frame passes through a multi-stage analysis pipeline: OCR text detection, banned word matching, and optionally ONNX-based image embedding comparison for threat classification. BLOCKED results trigger the blur FSM overlay and parent notification via FCM; SAFE results are stored locally in batches.

See System Implementation (Section 3.2) for the detailed AI analysis pipeline diagram with per-phase timing instrumentation.

4.4 Alert Generation Flow



4.5 Pairing Flow



See System Implementation (Section 4) for the step-by-step implementation details of the pairing flow.

5. Privacy and Ethical Considerations

5.1 Privacy by Design

The ANIS Child Application is architected with privacy as a foundational principle rather than an afterthought. Key design decisions include:

- On-device processing: All sensitive content analysis (OCR, image classification, banned word detection) is performed entirely on the device using ONNX Runtime and Google ML Kit. Raw screen captures, extracted text, and analysis images never leave the device.
- Data minimization: Analysis results transmitted to the backend are limited to non-image metadata: per-frame decisions (BLOCKED/SAFE), threat category scores, timestamps, and device performance metrics. This provides parents with behavioral visibility without exposing the actual content viewed by the child. Location coordinates and application package names are transmitted for their respective monitoring functions.
- No raw content transmission: The following data types are processed on-device only and are never transmitted to the backend or any external server:
- Screen capture bitmaps (Raw captured frames)
- Extracted OCR text (Detected text from screen captures)
- Banned word match details (Specific words detected)
- Model inference intermediate data (Embedding vectors, feature maps)
- Controlled data collection: The application only collects data explicitly required for its monitoring and enforcement functions. Telemetry data (location, apps) is collected at configurable intervals with clear user notification.
- Transparency by design: The persistent monitoring notification and the Home Screen status indicator provide clear, ongoing awareness that monitoring is active.

5.2 Child Privacy Protection

- Secure data handling: All stored data is protected through Android's sandboxed storage model. Captured images are stored in the application's internal storage directory, inaccessible to other applications.
- Limited retention: Location telemetry is automatically cleaned up after successful upload. Analysis results and captured images are retained only for the duration needed for session analysis and export.
- No unauthorized sharing: The application does not share collected data with any third-party services. All data is transmitted exclusively to the ANIS backend, which is controlled by the system administrators.
- Age-appropriate controls: The application provides age-appropriate privacy controls. Younger children cannot modify monitoring settings (PIN gated), while older children can view their own data through the Reward and Task screens.

5.3 Parent Transparency

- Clear policies: All monitoring policies and their implications are communicated to both the parent and the child during initial setup and when policies change.
- Consent mechanisms: Initial device pairing requires explicit parent consent through the parent application. The QR pairing process establishes a bilateral agreement between parent and child devices.
- Visible indicators: The monitoring status is visibly indicated on the child's device through a persistent notification and the Home Screen status display.
- Behavioral visibility: The parent application provides visibility into the child's digital behavior, including:
- Blocked content events with timestamps and threat categories (e.g., "Adult content blocked at 3:15 PM")
- Session summaries showing browsing patterns (total captures, blocked vs safe ratio)
- Behavioral trends over time (increasing or decreasing blocked event frequency)
- Device resource metrics during monitoring (battery drain, CPU usage)
- Application usage reports (time spent per app)

- Location history for geofencing awareness
- Reward activity
- No content preview: Parents can see that content was blocked and which category it fell into, but cannot view the actual screen capture or extracted text. This preserves the child's privacy while providing meaningful oversight.

5.4 Regulatory Compliance

The application is designed with reference to major child safety and data protection frameworks:

- COPPA (Children's Online Privacy Protection Act): The application limits data collection to what is reasonably necessary for its monitoring functions. Parental consent is required for data collection.
- GDPR-K (General Data Protection Regulation - Kids): Data minimization principles are applied. On-device processing reduces the scope of personal data processing. Parents have rights to access, rectify, and delete collected data through the parent application.
- Age-appropriate design: The user interface is designed to be understandable by children, with clear indicators of monitoring status and accessible explanations of why certain content is blocked.
- Data retention limits: Automatic cleanup of telemetry and analysis data ensures data is not retained beyond its operational necessity. The default retention period is 7 days for analysis data.

6. Benefits and Impact

6.1 Child Benefits

- Safer online experience: Real-time content protection actively identifies and blocks harmful content across multiple threat categories, providing immediate protection during device usage.
- Better digital habits: Screen time management and scheduled access periods encourage balanced device usage, reducing excessive or unhealthy screen time patterns.
- Educational engagement: The reward system provides structured incentives for educational content consumption, transforming passive screen time into active learning opportunities.
- Positive reinforcement: The reward system creates a positive feedback loop that encourages good digital behavior rather than relying solely on restriction and punishment.
- Transparency: The application provides clear indicators when monitoring is active and explains why specific content is blocked, helping children understand digital safety boundaries.

6.2 Parent Benefits

- Increased visibility: Real-time dashboards and periodic reports provide comprehensive insight into the child's digital activity, including application usage, location history, and content violations.
- Better control: Configurable policies allow parents to set appropriate boundaries for each child based on their age, maturity, and individual needs.
- Reduced supervision effort: Automated monitoring and enforcement reduce the need for constant manual supervision, while still providing immediate alerts for violations.
- Peace of mind: Real-time content protection and location tracking provide confidence that the child is safe both online and offline.
- Positive behavior tools: The reward and task systems provide tools for encouraging positive behavior rather than only restricting negative behavior.

6.3 Platform Benefits

- Balanced safety and privacy: On-device AI processing ensures content is analyzed without transmitting sensitive data to external servers, balancing safety requirements with privacy considerations.
- Scalable architecture: The layered architecture with offline-first design ensures the system continues functioning even without network connectivity, scaling from individual families to large deployments.

- AI-assisted protection: On-device machine learning provides adaptive, intelligent content moderation that improves with model updates without requiring server-side processing.
- Native performance: The native Android implementation provides reliable background execution, efficient memory usage, and predictable latency for time-sensitive content moderation.
- Comprehensive integration: Accessibility Service integration provides system-wide monitoring capabilities that go beyond what application-level monitoring can achieve.

System Implementation

1. Device Monitoring

1.1 Location Tracking

The application collects location data using the FusedLocationProviderClient (Google Play Services) with a periodic WorkManager task ensuring hourly uploads. Location data is persisted locally in a Room database (LocationTelemetryEntity) with an isSent flag to support offline operation.

Location collection flow:

- TelemetryManager registers a location request via FusedLocationProviderClient with a minimum interval of 30 minutes and minimum distance threshold of 100 meters.
- Collected locations are inserted into the Room database with isSent = false.
- A periodic WorkManager task (LocationTelemetryWorker) runs every hour and:
- Reads all unsent locations ordered by timestamp.
- Uploads each location via the backend API (POST locations/telemetry/{childId}).
- Marks successfully uploaded locations as sent.
- Deletes sent records to manage database size.
- Failed uploads are retried with exponential backoff (base 30s, max 60s, up to 5 retries).
- An FCM-triggered immediate sync can be invoked by sending a silent push notification with sync_locations or trigger_sync data payloads.

1.2 Installed Applications Monitoring

The AppsRepository queries the Android PackageManager for all non-system launcher applications on the device. The application list is sent to the backend via POST apps/add-bulk. This enables the parent to view installed apps and configure per-app restrictions.

1.3 Session Monitoring

During active AI monitoring sessions, the system tracks device resource metrics:

- Battery level: Percentage at session start and end via BatteryManager.
- Battery charging status: Whether the device was charging during the session.
- CPU time: Extracted from /proc/self/stat.
- RAM PSS: Measured via Debug.MemoryInfo at session boundaries.

Per-capture analytics include timestamps, decision outcome, OCR latency, and ONNX inference latency.

2. Screen and App Usage Time Management

The system manages device usage time at two levels: overall screen time for the entire device and per-app restrictions configured by the parent.

2.1 Application Management

- Application discovery: Periodic scanning of installed packages via PackageManager.
- Categorization: Applications can be categorized (Education, Games, Social, Productivity) based on package metadata and configurable rules.
- Application blocking: Restricted applications can be blocked via system overlay interception and Accessibility Service navigation control.
- Time limits: Per-application daily usage limits are enforced at the system level by tracking foreground time via UsageStatsManager and interrupting access when limits are reached.

2.2 Screen and App Usage Time

Overall Screen Time:

- Daily limits: Total device usage is capped at a parent-configured daily limit.
- Scheduled access: Specific time windows (e.g., study hours, bedtime) restrict device usage.
- Temporary restrictions: Parents can impose immediate short-term restrictions through the parent application.
- Automatic enforcement: Limits are enforced locally via Accessibility Service and system overlay mechanisms. No backend connectivity is required for enforcement.
- Usage recovery: Children can earn additional screen time through successful task completion, with the earned time immediately applied to the current day's allowance.

Per-App Restrictions:

Parents can create individual app restrictions through the parent application for specific applications on the child's device:

- Blocked apps: Selected applications can be completely blocked during configured time windows.
- Time-limited apps: Specific apps can be assigned individual daily usage limits, independent of the overall device limit.
- Allow-listed apps: Certain apps (e.g., educational or communication tools) can be exempted from restrictions.
- App list synchronization: The child device sends the installed application list to the backend via POST apps/add-bulk, and the parent selects which apps to restrict through the parent application interface.
- Enforcement: Blocked apps are detected via the Accessibility Service and prevented from launching or are immediately backgrounded when opened.
- Live sync: When the parent modifies app restrictions through the parent application, an FCM push notification is sent to the child device to trigger an immediate sync of the updated policy. The child device fetches the latest restrictions from the backend and applies them without requiring a restart or manual refresh.

2.3 Restriction Enforcement

Restrictions are enforced locally through multiple mechanisms:

- Real-time enforcement: The foreground monitoring service evaluates each AI analysis result against active policies. BLOCKED content triggers immediate overlay and notification actions.
- Accessibility Service: Detects application launches and navigation events, enabling policy-based interception.
- System overlay blocking: When an application is restricted, a system overlay prevents interaction with the app.

3. Content Protection System

The Content Protection System is the core AI-powered feature of the application. It performs real-time on-device content moderation using a CLIP-based vision model running via ONNX Runtime, combined with Google ML Kit for optical character recognition.

3.1 Architecture Overview

The system is composed of five coordinated components:

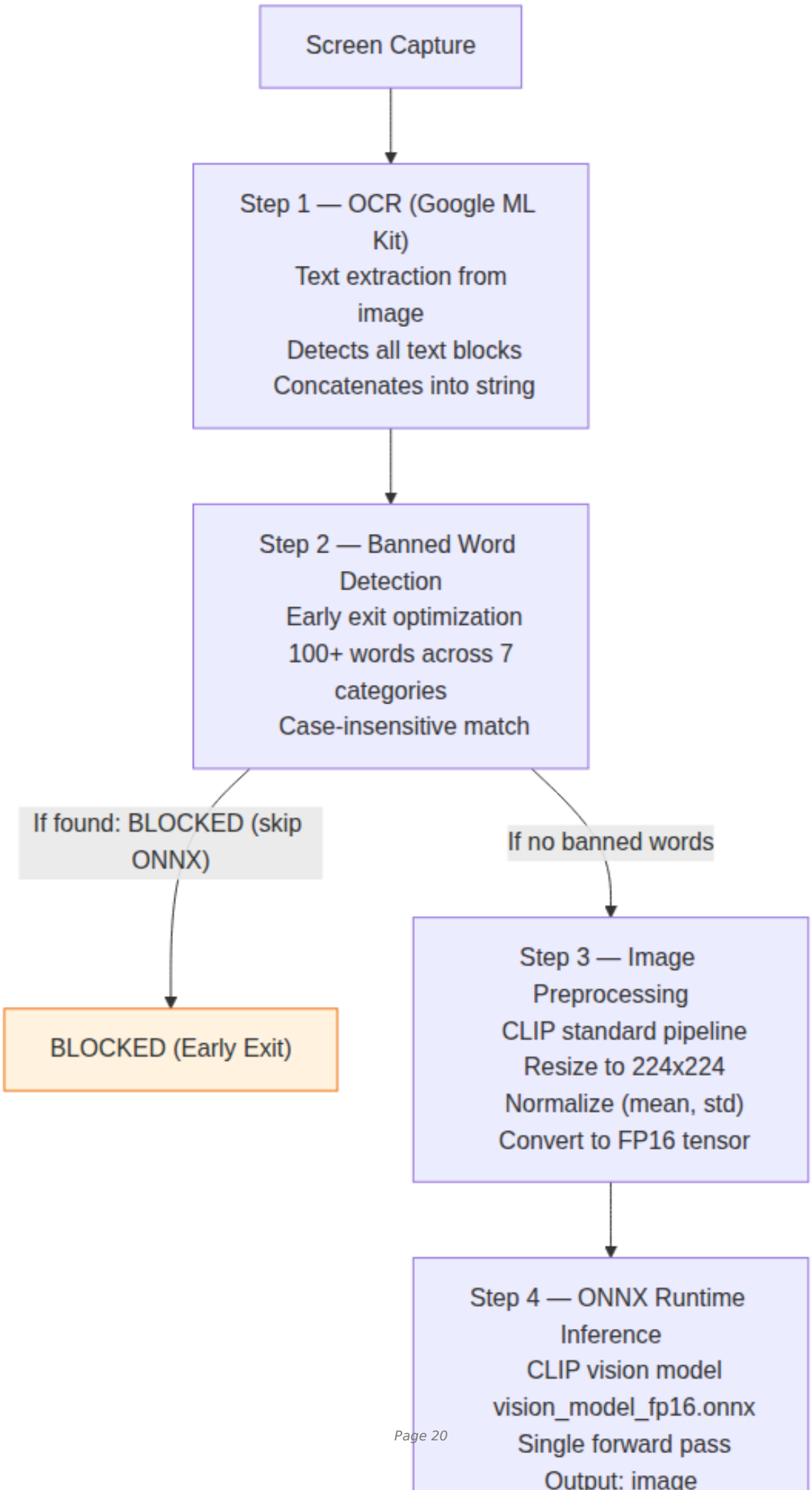
Component	Responsibility
AiAnalyzer	Core inference engine: ONNX model loading, image preprocessing, embe
SessionManager	Session lifecycle orchestration, configuration, device stats monitoring
SessionCaptureService	Foreground service: MediaProjection capture loop, VirtualDisplay manage

BlurOverlayManager	BroadcastReceiver-based show/hide of frosted-glass overlay
BlurNotificationManager	High-priority notifications for blocked content alerts

These components are wired together via Hilt dependency injection, with the SessionManager acting as the central coordinator.

3.2 AI Analysis Pipeline (AiAnalyzer)

The analysis of each captured frame follows a strict sequential pipeline with early-exit optimization:


Screen Capture
Step 1 — OCR (Google ML Kit)
Text extraction from image
Detects all text blocks
Concatenates into string
Step 2 — Banned Word Detection
Early exit optimization
100+ words across 7 categories
Case-insensitive match
If found: BLOCKED (skip ONNX)
If no banned words
BLOCKED (Early Exit)
Step 3 — Image Preprocessing
CLIP standard pipeline
Resize to 224x224
Normalize (mean, std)
Convert to FP16 tensor
Step 4 — ONNX Runtime Inference
CLIP vision model
vision_model_fp16.onnx
Single forward pass
Output: image

OCR and Banned Word Detection

Google ML Kit Text Recognition scans each frame for text content. The extracted text is checked against a banned word list organized into 7 categories:

Category	Examples
Profanity	Common swear words and obscenities
Violence	Words related to fighting, weapons, harm
Adult	Sexually explicit or suggestive terms
Drugs	Substance-related terminology
Weapons	Firearm, knife, explosive references
Hate	Discriminatory or derogatory language
Gambling	Betting, casino-related terms

Detection is case-insensitive. If any banned word is found, the pipeline exits early with a BLOCKED decision, avoiding the computationally expensive ONNX inference step entirely.

Image Preprocessing

Before ONNX inference, the captured bitmap undergoes CLIP-standard preprocessing:

- Resize to 224x224 pixels (CLIP vision encoder input size).
- Normalize using CLIP mean and standard deviation values.
- Convert to FP16 normalized tensor via buffer allocation and memory mapping.

ONNX Runtime Inference

The preprocessed tensor is fed into the CLIP-based vision model:

- Model: vision_model_fp16.onnx, a CLIP ViT variant with FP16 quantization.
- Runtime: ONNX Runtime Android (OrtSession).
- Output: A fixed-dimensional embedding vector representing the image content.
- Performance: Approximately 372 ms average inference time on modern Android devices.

Embedding Comparison and Threat Classification

The extracted embedding is compared against pre-computed threat embeddings loaded from saved_embeddings.json:

- L2 Normalization of both image embedding and threat embeddings.
- Cosine similarity via dot product of normalized vectors.
- Logit scaling using temperature scaling factor exp(4.60517).
- Threshold comparison against pre-computed threat embeddings using default thresholds: Adult = 0.35, Violence = 0.40, Other = 0.40.

If any threat score exceeds its threshold, the content is classified as BLOCKED. Otherwise, it is classified as SAFE.

Performance Instrumentation

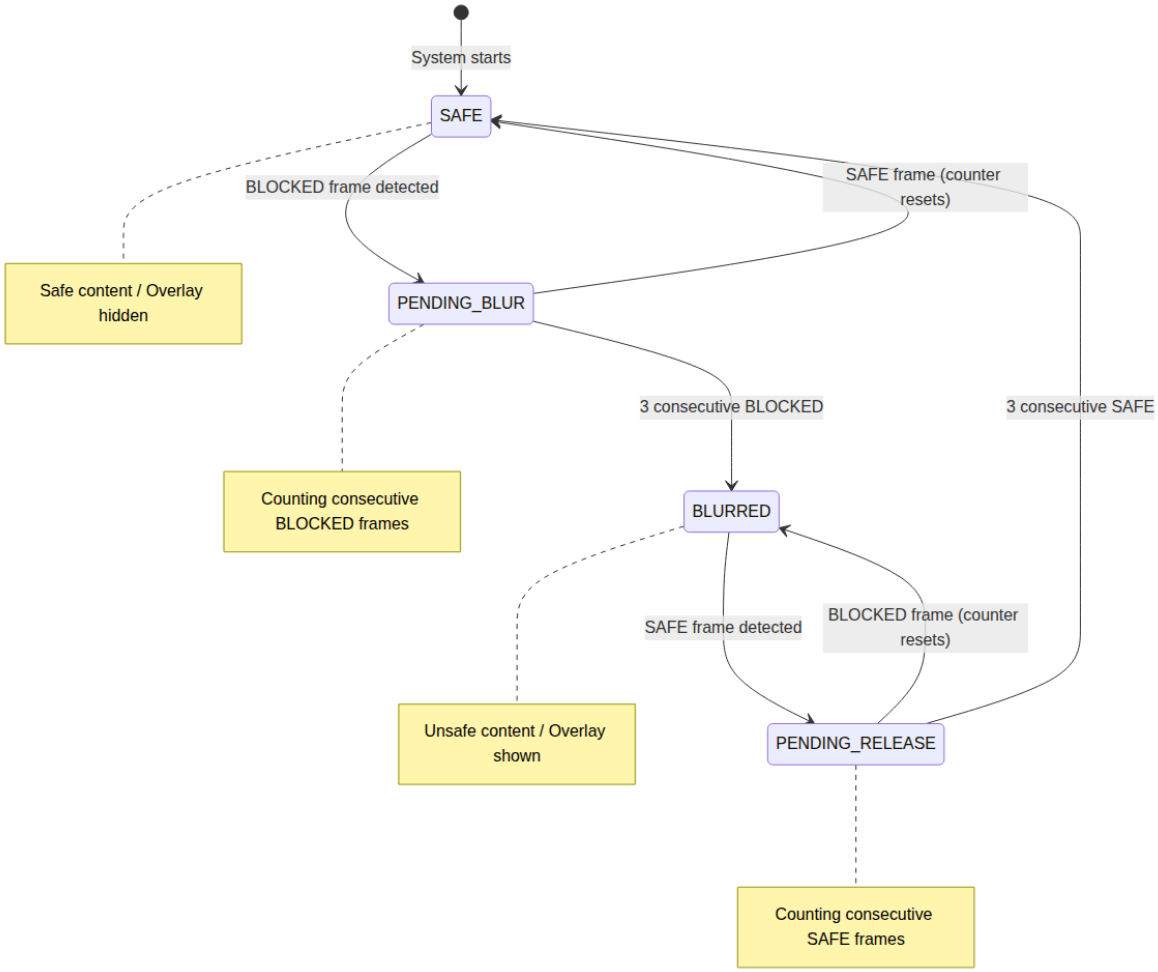
Every phase of the pipeline is instrumented with timing measurements:

Phase	Timing Variable	Typical Duration
OCR	ocrTimeMs	~176 ms
Preprocessing	preprocessTimeMs	~97 ms
ONNX Inference	onnxTimeMs	~372 ms
Total		~645 ms

These timings are logged and stored per-capture for analytics and performance monitoring.

3.3 Blur Overlay Finite State Machine

The blur overlay is not toggled on every single BLOCKED frame. A finite state machine prevents flickering from transient content:



(ده زی باتاع فتوح الللی بيقول UI Blur stab. logic)

- SAFE -> PENDING_BLUR: First BLOCKED frame detected; start counting consecutive violations.
- PENDING_BLUR -> BLURRED: Configurable threshold reached (default: 3 consecutive BLOCKED frames); show overlay.
- BLURRED -> PENDING_RELEASE: First SAFE frame while blurred.
- PENDING_RELEASE -> SAFE: Configurable consecutive SAFE frames (default: 3); hide overlay.
- PENDING_BLUR/PENDING_RELEASE: State resets on opposite decision.

Default thresholds:

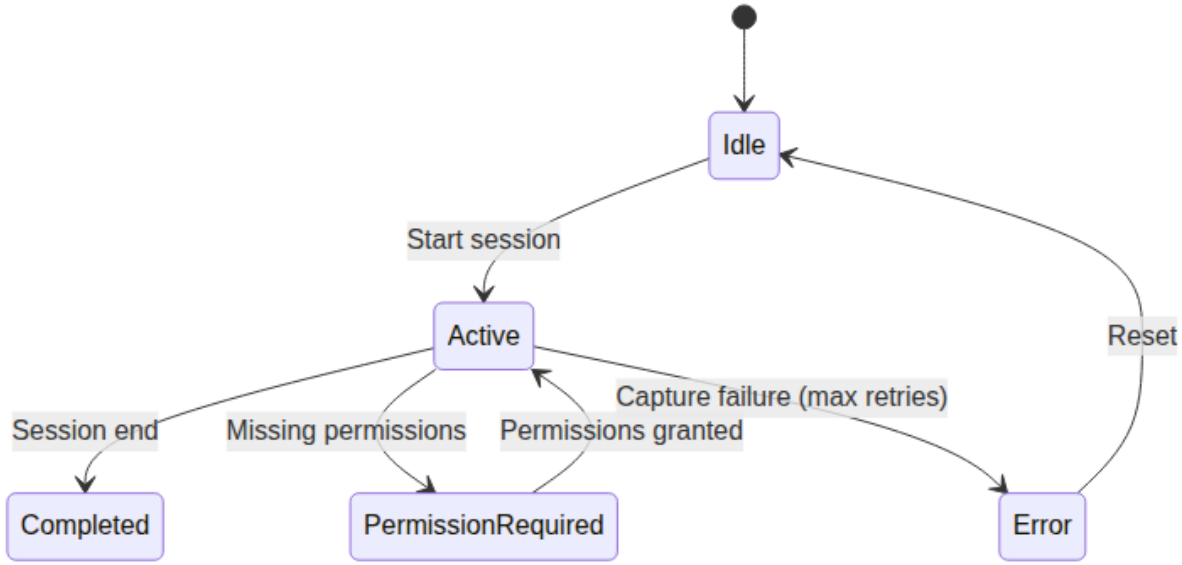
- blurTriggerThreshold: 3 consecutive unsafe frames.
- blurReleaseThreshold: 3 consecutive safe frames.

Figure 6.1: Frosted-glass blur overlay activated when unsafe content is detected. The overlay blocks the app content until sufficient safe frames are observed.

3.4 Session Lifecycle

Sessions organize continuous monitoring periods with full lifecycle tracking.

Session States



Session Configuration

Parameter	Default	Range
sessionIntervalMs	1000 ms	500-5000 ms
blurTriggerThreshold	3 frames	1-10

Device Resource Monitoring

Each session tracks device-level metrics at start and end: battery level via BatteryManager, CPU time from /proc/self/stat, and RAM PSS via Debug.MemoryInfo.

Capture Loop

```
while (session is Active) {
    1. Capture frame via MediaProjection + ImageReader
    2. Extract bitmap from ImageReader
    3. Run AiAnalyzer.analyzeImage(bitmap)
    4. Apply blur FSM decision
    5. Store analysis result (batched for SAFE, immediate for BLOCKED)
    6. Wait for configured interval
}
```

Batched vs Immediate Writes

To minimize database I/O:

- SAFE results: Batched, written to Room DB every 10 results or every 5 seconds.
- BLOCKED results: Written immediately to ensure no data loss on crash.

Fault Tolerance

- Auto-restart: Up to 5 consecutive restart attempts on capture failure.
- Exponential backoff: Increasing delay between restart attempts.
- Max retry cap: After 5 failures, session enters Error state.

3.5 Database Schema

sessions table

Column	Type	Description
id	Long (PK, auto)	Session identifier
startTime	Long	Epoch millis of session start
endTime	Long (nullable)	Epoch millis of session end

status	String	Idle, Active, Completed, Error
intervalMs	Long	Capture interval in ms
totalCaptures	Int	Total frames processed
blockedCount	Int	Number of BLOCKED decisions
safeCount	Int	Number of SAFE decisions
batteryStart	Int	Battery % at session start
batteryEnd	Int (nullable)	Battery % at session end
batteryCharging	Boolean	Charging state
cpuTimeMs	Long	Total CPU time
cpuUsagePercent	Double	CPU usage percentage
ramPssMb	Double	RAM PSS in MB

analysis_results table

Column	Type	Description
id	Long (PK, auto)	Result identifier
sessionId	Long (FK -> sessions)	Parent session
timestamp	Long	Capture timestamp
analysisResult	String	Safe or Blocked
decision	String	BLOCKED or SAFE
ocrTimeMs	Long	OCR duration
onnxTimeMs	Long	ONNX inference duration
threatDetails	String (JSON)	Per-category threat scores
imagePath	String (nullable)	Path to captured image file

3.6 Behavior Reporting and Parent Preview

Analysis results from each monitoring session are transmitted to the backend to provide parents with visibility into their child's digital behavior. This enables the parent application to display behavior reports and trends over time.

Data transmitted per analysis result:

Field	Description	Privacy Status
sessionId	Session identifier	Non-identifying
timestamp	Capture timestamp	Required for timeline
decision	BLOCKED or SAFE	Behavior indicator
analysisResult	Blocked category (if applicable)	Behavior indicator
ocrTimeMs	OCR processing time	Performance metric
onnxTimeMs	ONNX inference time	Performance metric
threatDetails	Per-category threat scores (JSON)	Behavior indicator

Voting System for Parent Review

A subset of captured frames is selected throughout the day and sent to the parent via API for manual review and voting. This allows the parent to validate or override the on-device AI model's decisions, creating a feedback loop that improves local model accuracy over time.

- Selected frames are sampled at a configurable interval during active monitoring sessions.
- Each frame is sent to the backend with the on-device decision (BLOCKED / SAFE) and threat scores.

- The parent reviews the image in the parent application and casts a vote (BLOCKED / SAFE) on each submission.
- Votes are sent back to the child device and used as labeled data for local model refinement.
- Detailed implementation of the voting system, including the local model update pipeline, is described in the ANIS AI Services documentation.

Not transmitted (unless selected by the voting system):

- Captured images (stored locally only, used for on-device analysis)
- Extracted OCR text (used only for on-device banned word detection)
- Application-specific content or personal data

Parent visibility:

The parent application can view:

- Blocked content events with timestamps and threat categories
- Session summaries (total captures, blocked count, safe count)
- Behavioral trends over time (blocked event frequency, safe browsing duration)
- Device resource metrics during monitoring sessions (battery, CPU, RAM)
- Voting queue with images flagged by the sampling system for manual review

This approach balances the parent's need for behavioral insight with the child's privacy: detailed content analysis stays on-device, while aggregated behavior indicators and a sampled subset of frames are reported for parental oversight.

3.7 Export System

Individual session data can be exported for analysis:

- XLSX export: Session summary sheet and detailed analysis results sheet.
- ZIP export: XLSX file bundled with all captured images from the session.
- All sessions export: Aggregated XLSX across all sessions.

Export uses FastExcel for XLSX generation and Android's ZipOutputStream for archive creation.

3.8 Protected Content Overlay

When blocked content is detected and the FSM triggers the BLURRED state:

- Frosted glass overlay: White background with 94% opacity, system overlay using TYPE_APPLICATION_OVERLAY.
- BroadcastReceiver mechanism: BlurOverlayManager sends and receives broadcasts to show or hide the overlay.
- High-priority notification: BlurNotificationManager displays a heads-up notification on blocked content with vibration alert.
- Auto-recovery: After blurReleaseThreshold consecutive safe frames, overlay is dismissed automatically.

3.9 Permission Requirements

The content protection system requires three Android runtime permissions:

Permission	Purpose
MediaProjection	Screen capture for analysis
SYSTEM_ALERT_WINDOW	Blur overlay display
POST_NOTIFICATIONS	Blocked content alerts (Android 13+)

Permission flow:

- Application checks if all permissions are granted.
- If missing, session transitions to PermissionRequired state.

- User is navigated to Permissions Screen.
- After granting, session resumes automatically.

4. Device Pairing

The device pairing process establishes a secure association between the child device and the parent account. See System Architecture and Design (Section 4.5) for the sequence diagram of the full pairing flow.

- The parent generates a QR code through the parent application containing a JSON payload with action, childId, and token.
- The child application uses CameraX with ML Kit Barcode Scanning to scan the QR code.
- The PairingViewModel parses the QR data and constructs a PairingRequest with the device ID, device name, and FCM token.
- The AuthRepository sends the pairing request to POST children/pair.
- On success, the backend returns an access token which is stored in EncryptedSharedPreferences.
- The child profile (name, age, email, avatar URL) is fetched via GET children/me.
- The application transitions from the unpaired state to the paired state.

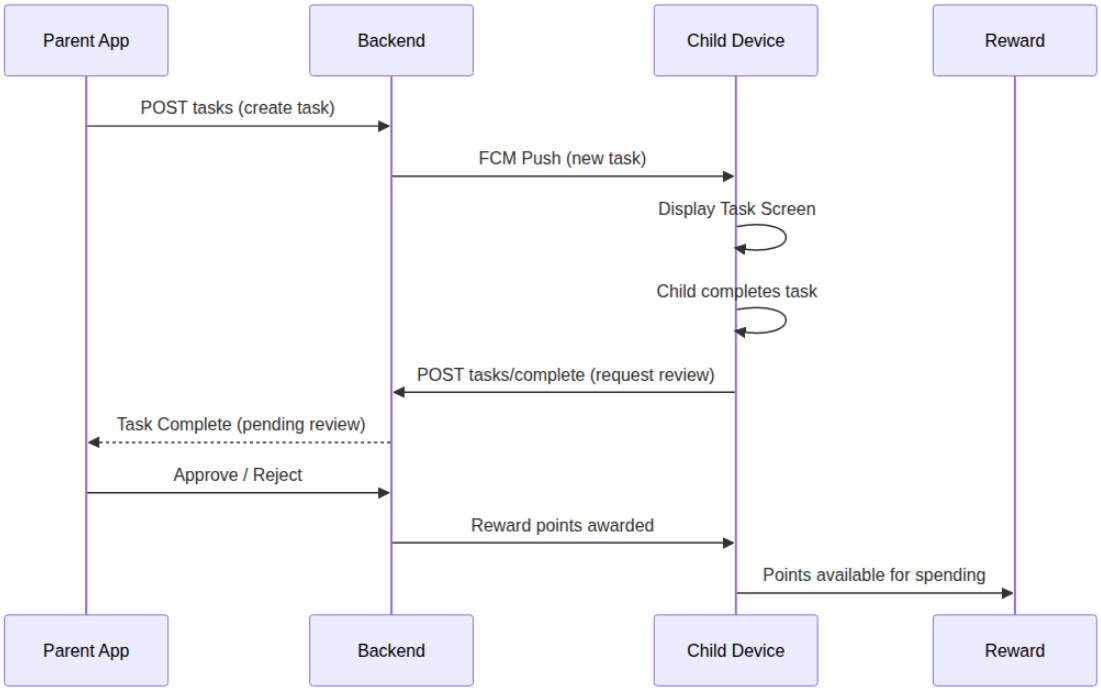
Figure 4.1: QR code scanning screen during device pairing.

5. Task System

5.1 Overview

The Task System provides a structured mechanism for parents to assign activities and responsibilities through the parent application. Children complete tasks to earn reward points, which they can then spend on rewards via the Reward Screen. This creates a positive reinforcement loop that encourages productive behavior and responsibility.

5.2 Task Lifecycle



- Task creation: A parent creates a task through the parent application, specifying title, description, due date, and reward point value.
- Delivery: The backend stores the task assignment and sends a push notification to the child device via FCM.

- Display: The child opens the Task Screen to view all assigned tasks with their point values and due dates.
- Completion: The child completes the task and submits a completion request (POST tasks/complete) for parent review.
- Review: The parent reviews the submission through the parent application and approves or rejects it.
- Reward: Upon approval, reward points are credited to the child's balance, available for spending on the Reward Screen.

5.3 Task Screen

The Task Screen is accessible from the Home Screen without a PIN, ensuring children can freely view and manage their responsibilities. The screen presents:

- Task title and description with clear instructions.
- Reward point value displayed for each task.
- Due date and remaining time indicator.
- Completion status (pending, submitted, approved, rejected).
- Submit button that triggers a completion review request.

Figure 5.1: Task screen displaying parent-assigned tasks with point values, due dates, and completion status.

6. Reward Management System

6.1 Objectives

The Reward Management System reinforces positive behavior and encourages healthy digital habits through a structured incentive framework. The system aims to:

- Create a positive feedback loop linking good digital behavior with tangible rewards.
- Provide parents with a flexible tool for behavior management.
- Give children visibility into the consequences of their digital choices.
- Integrate with the content protection system to reward sustained safe behavior.
- Provide a redemption marketplace where children can spend earned points on rewards of their choice.

6.2 Reward Points Economy

The reward system operates on a points-based economy:

- Earning points: Children earn reward points primarily by completing parent-assigned tasks. Each task has a point value set by the parent during creation.
- Bonus points: Additional points can be earned through sustained safe browsing periods (no content violations) and surprise rewards issued by the parent.
- Spending points: Children redeem their accumulated points on the Reward Screen to purchase rewards configured by the parent.
- Point deductions: Policy violations or incomplete tasks may result in point deductions at the parent's discretion.

6.3 Reward Types

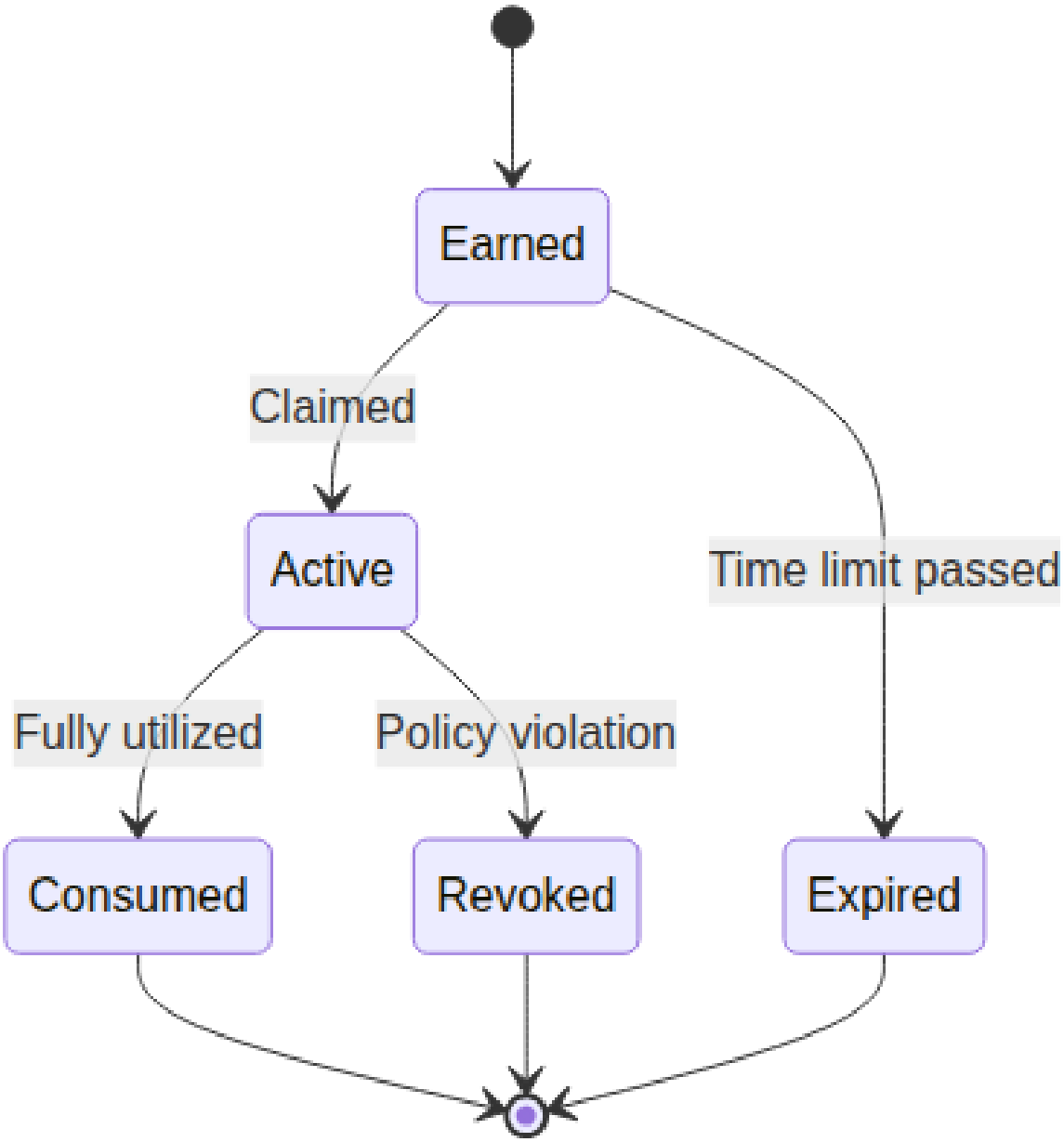
Reward Type	Description	Parent Configurable
Additional Screen Time	Extra minutes added to daily allowance	Yes (duration)
Temporary Privilege Unlocks	Access to restricted apps for a limited perio	Yes (apps, duration)
Achievement Badges	Milestone-based recognition for sustained g	Yes (criteria)
Custom Rewards	Parent-defined rewards (e.g., "Choose week	Yes (description)

6.4 Dynamic Reward Adjustment

Parents can dynamically adjust the reward system through the parent application:

- Modifications: Add, remove, or modify reward types and values at any time.
- Reduction: Reward values can be reduced for policy violations.
- Enhancement: Sustained good behavior can trigger reward value increases.
- Surprise rewards: Parents can issue unscheduled rewards for exceptional behavior.

6.5 Reward Lifecycle



State	Description
Earned	Reward has been earned but not yet activated or claimed
Active	Reward is currently in effect (e.g., extra time being used)
Consumed	Reward has been fully utilized
Expired	Time-limited reward that was not activated before expiration
Revoked	Reward cancelled due to policy violation

6.6 Integration with Content Protection

The reward system is directly integrated with the content protection system:

- Positive behavior: Periods of uninterrupted safe browsing (no blocked content) contribute to reward accumulation.
- Violations: Blocked content events can reduce or revoke accumulated rewards based on severity and frequency.
- Transparency: The child can see the direct correlation between their browsing behavior and their reward balance through the Reward Screen.
- Notification: Both positive reward events and reward reductions trigger notifications on the child's device.

6.7 Reward Screen

The Reward Screen is accessible from the Home Screen without a PIN, allowing children to freely monitor their progress and spend earned points. The screen displays:

- Current reward point balance and equivalent value.
- Available rewards catalog showing each reward's point cost (configured by the parent).
- Purchase button to redeem points for a selected reward.
- Reward history with timestamps and descriptions.
- Pending rewards awaiting parent approval.
- Expired or revoked rewards with explanations.
- Behavior score or progress indicator toward next reward milestone.

Figure 6.1: Reward overview screen showing balance, history, and pending approvals.

7. Android Components

7.1 Activities

MainActivity is the single activity that hosts the entire Compose UI. It extends ComponentActivity and acts as the entry point for all user interactions.

Responsibilities:

- Manages runtime permission requests (Camera, Location, MediaProjection, Overlay, Notifications).
- Creates and initializes the PreferenceManager, TelemetryManager, and LogManager.
- Hosts the Compose content tree.
- Controls top-level navigation between the unpaired state (PairingScreen) and paired state (HomeScreen with access to Settings, Reward, and Task screens) using a boolean isLoggedIn state variable.
- Handles the MediaProjection permission result via onActivityResult.

7.2 Jetpack Compose Screens

Screen	Purpose	PIN Required
PairingScreen	QR code scanning via CameraX and ML Kit E	No
HomeScreen	Welcome screen displaying child name, mor	No
PinScreen	Numeric PIN entry for parent gate to unlock	N/A (gate itself)
SettingsScreen	Monitoring toggle, dark mode, permission st	Yes
TaskScreen	Task list and completion tracking	No
RewardScreen	Reward balance display, reward history, beh	No
BlockedContentScreen	Notification and details of blocked content e	No

Figure 7.1: Home screen showing child greeting, monitoring status, recent activity summary, and navigation tiles.

Figure 7.2: Settings screen with monitoring toggle, permission indicators, and parent gate (PIN-protected).

7.3 ViewModels

ViewModel	Key State (StateFlow)	Key Actions
PairingViewModel	PairingUiState (Scanning, Loading, Success,	parseQrData(), pairDevice()
HomeViewModel	Loading states for location, apps, profile	sendCurrentLocation(), sendInstalledApps(), fetchC
PinViewModel	PinUiState (entry, validation, lockout)	validatePin(), setPin(), clearAttempts()
SettingsViewModel	MonitoringEnabled, DarkMode, PermissionSt	toggleMonitoring(), toggleDarkMode(), checkPermis
TaskViewModel	TaskList, completion status	fetchTasks(), completeTask()
RewardViewModel	RewardBalance, RewardHistory	claimReward(), fetchHistory()

7.4 Foreground Services

MonitoringService (Capture Service):

- Runs as a foreground service with a persistent notification.
- Manages the MediaProjection session and VirtualDisplay creation.
- Implements the capture loop with configurable interval (500-5000 ms).
- Coordinates with AiAnalyzer for per-frame content analysis.
- Handles the blur overlay FSM transitions.
- Implements auto-restart with exponential backoff on failure.

LocationService:

- Manages periodic location collection via FusedLocationProviderClient.
- Stores collected locations to Room database for batched upload.

7.5 Accessibility Service

The Accessibility Service (android.accessibilityservice.AccessibilityService) provides system-wide monitoring capabilities:

- App usage tracking: Detects the current foreground application and tracks time spent per application.
- Navigation interception: Intercepts system navigation events for restriction enforcement.
- App launch detection: Triggers policy evaluation when a new application is opened.
- Persistent background operation: The service continues running even when the application UI is not visible.
- Configuration: Declared in AndroidManifest.xml with BIND_ACCESSIBILITY_SERVICE permission and configured via accessibility-service.xml for event types and package tracking.

7.6 MediaProjection Service

The MediaProjection Service captures screen content for AI analysis:

- Uses MediaProjectionManager to create a MediaProjection session.
- Creates a VirtualDisplay with an ImageReader surface (PixelFormat.RGBA_8888).
- The ImageReader acquires frames at the configured capture interval.
- Captured Image objects are converted to Bitmaps for analysis.
- The service is started via an Intent obtained from createScreenCaptureIntent() with result handling in MainActivity.

7.7 Broadcast Receivers

Receiver	Purpose
BootReceiver	Restarts monitoring and location services after device reboot
ConnectivityReceiver	Triggers pending data synchronization when network becomes available
ScreenStateReceiver	Pauses monitoring when screen is off, resumes when screen is on

BlurOverlayReceiver	Internal communication between BlurOverlayManager and UI for show/hi

7.8 WorkManager Tasks

Worker	Schedule	Purpose
LocationTelemetryWorker	Periodic (every 1 hour)	Uploads unsent location data with exponential back
LocationImmediateSync	One-shot (FCM triggered)	Immediate location upload on push notification
AppsSyncWorker	Periodic (daily)	Reports installed applications list
PolicySyncWorker	Periodic + FCM triggered	Fetches latest restriction policies from backend
RewardSyncWorker	Periodic + event driven	Synchronizes reward state

Workers use the following constraints:

- NetworkType.CONNECTED (do not run without internet).
- Exponential backoff with base 30 seconds, max 60 seconds, up to 5 retries.

7.9 Notifications

Notification Channel	Importance	Purpose
Monitoring Service	Low (priority)	Persistent foreground service indicator
Blocked Content	High	Heads-up alert with vibration for blocked content
Task Assigned	Default	New task assignment notification
Reward Earned	Default	Positive behavior reward notification
Reward Expiring	Default	Time-limited reward about to expire
Monitoring Status	Low	Status updates and information

The notification system covers multiple event types:

- Blocked content alerts: High-priority notification with vibration when inappropriate content is detected.
- Parent instructions: FCM push notifications deliver real-time instructions from parents.
- Task notifications: Alerts when new tasks are assigned by the parent.
- Reward notifications: Notifications when rewards are earned, claimed, or expiring.
- Policy sync notifications: Alerts when app restrictions or limits are updated remotely.
- Monitoring status: Persistent foreground service notification indicating that monitoring is active.

7.10 Local Storage Components

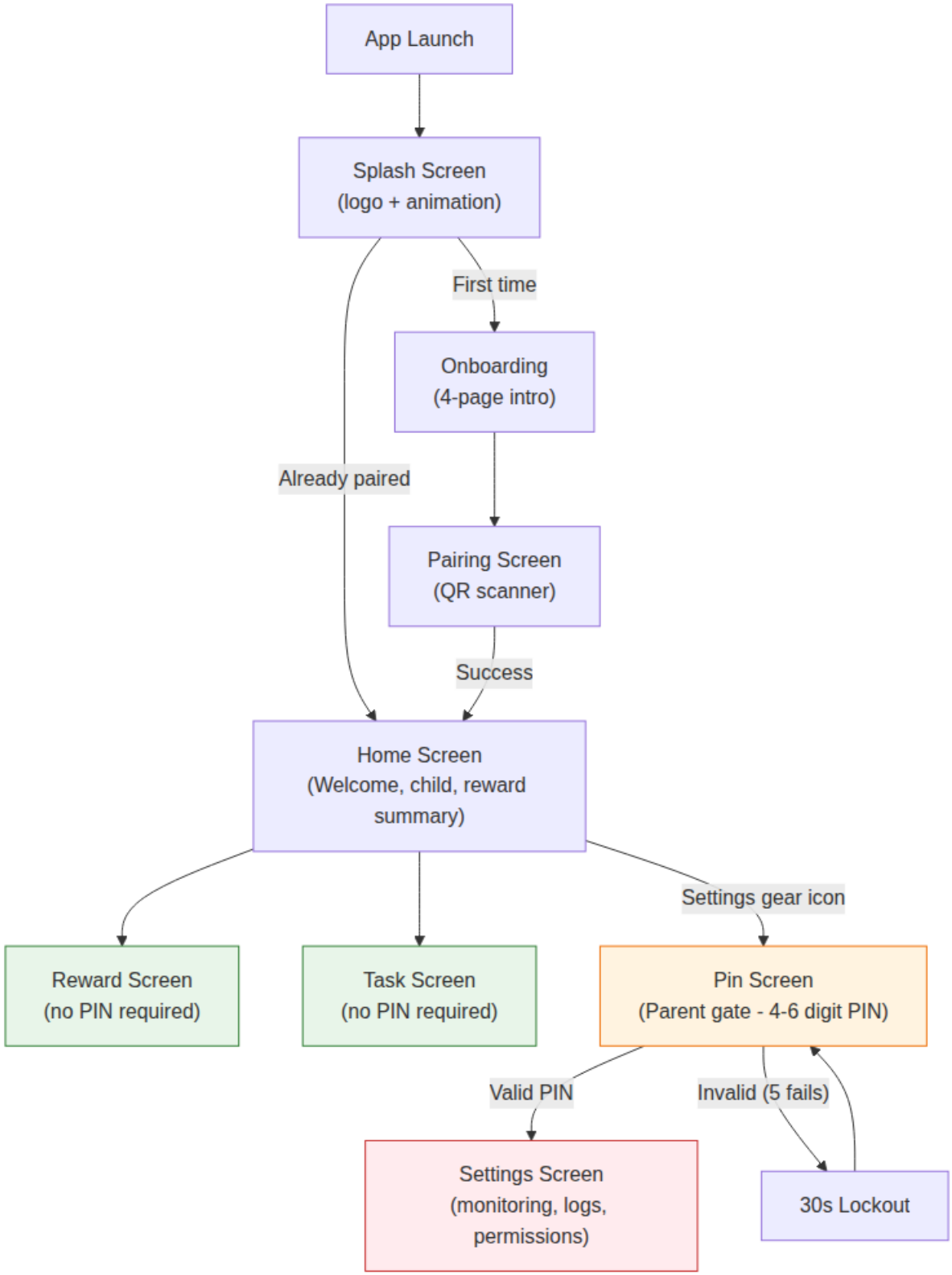
Storage	Technology	Data Stored
Room Database (anis_database)	SQLite + Room	Location telemetry, sessions, analysis results, rewa
EncryptedSharedPreferences	AES-256-GCM	Authentication token, FCM token, child profile, PIN
DataStore (Preferences)	Jetpack DataStore	Theme preference, monitoring toggle state, sessio
Internal Storage	App-specific filesystem	Captured analysis images, export archives

Room Database Entities:

- LocationTelemetryEntity: id, latitude, longitude, timestamp, isSent
- SessionEntity: id, startTime, endTime, status, intervalMs, totalCaptures, blockedCount, safeCount, battery metrics, CPU metrics, RAM metrics
- AnalysisResultEntity: id, sessionId (FK), timestamp, analysisResult, decision, ocrTimeMs, onnxTimeMs, threatDetails (JSON), imagePath
- RewardEntity: id, type, value, state (earned/active/consumed/expired/revoked), earnedAt, expiresAt
- TaskEntity: id, title, description, dueDate, completedAt, rewardValue

7.11 UI/UX Design

Screen Navigation Flow



Design Principles

- Material 3 Design: Full Material You theming with dynamic color support on Android 12+.
- Dark/Light Mode: System-aware with manual toggle in Settings. Custom color palettes for both modes.

- Single Activity Architecture: One MainActivity hosts all Compose screens; navigation controlled by sealed class state.
- Permission-Centric Flow: Application guides user through permission grants (Camera, Location, MediaProjection, Overlay, Notifications) before enabling core features.
- Monitoring Status Indicator: Persistent notification and Settings toggle show current monitoring state.
- Accessibility: Content descriptions on all UI elements; large touch targets; high-contrast mode support.
- Offline-Aware UI: UI elements gracefully handle offline states with appropriate messaging and disabled states for backend-dependent actions.

8. Communication and Synchronization

8.1 Backend Communication

The application communicates with the ANIS backend through a RESTful API over HTTPS. Communication is handled by Retrofit 2.9.0 with OkHttp 4.12.0 and kotlinx.serialization.

Network Stack Architecture:

```
+------------------+
|   ApiService     |  Retrofit interface (endpoint definitions)
+--------+---------+
         |
+--------+---------+
|   NetworkProvider |  Singleton: OkHttpClient + Retrofit builder
+--------+---------+
         |
+--------+---------+
|   OkHttpClient    |  Interceptors: AuthInterceptor,
|                   |  AppLoggingInterceptor, HttpLoggingInterceptor
+--------+---------+
         |
+--------+---------+
|   TLS 1.3         |  Certificate pinning (optional)
+------------------+
```

Base Configuration:

Parameter	Value
Base URL	https://api.anis.solutions/api/v1/
Connect Timeout	30 seconds
Read Timeout	30 seconds
Write Timeout	30 seconds
Content Type	application/json
Serialization	kotlinx.serialization JSON

Interceptors:

- AuthInterceptor: Reads the access token from EncryptedSharedPreferences and injects an Authorization: Bearer <token> header into every request. If no token is present, the request proceeds without authentication.
- AppLoggingInterceptor: Logs every HTTP request (method, path, body) and response (status code, elapsed time, body) via LogManager for in-app debugging.
- HttpLoggingInterceptor (OkHttp): Debug-level logging for development builds.

8.2 Behavior Data Synchronization

Analysis results from content monitoring sessions are transmitted to the backend for parent preview. The synchronization strategy distinguishes between safe and blocked events:

- BLOCKED results: Transmitted immediately via API call when connectivity is available, ensuring parents receive real-time alerts about policy violations.
- SAFE results: Batched locally and transmitted periodically (every 10 results or every 5 seconds), reducing network overhead while maintaining parent visibility into browsing patterns.
- Session summaries: At session completion, aggregated data (total captures, blocked count, safe count, device metrics, session duration) is uploaded to provide a complete behavior picture.
- Offline handling: All results are persisted locally with transmission status tracking. When connectivity is restored, pending results are uploaded in priority order (BLOCKED first, then SAFE batches, then session summaries).

8.3 Real-Time Synchronization

Firebase Cloud Messaging provides real-time push notification capabilities:

- FCM Token Registration: On first launch and on token refresh (onNewToken), the application registers its FCM token with the backend via POST children/fcm-token.
- Silent Push Notifications: The backend can send data-only push messages to trigger immediate actions without user-visible notifications:
- sync_locations: Triggers immediate location telemetry upload.
- trigger_sync: Triggers full data synchronization.
- sync_behavior: Triggers immediate upload of pending analysis results.
- Policy Updates: Policy changes made by the parent are propagated to the child device via push notification, triggering PolicySyncWorker for immediate enforcement.
- Reward Updates: Reward changes and assignment notifications are delivered via push.

8.4 API Endpoints

Method	Endpoint	Purpose
POST	children/pair	Device pairing with QR authentication
POST	children/fcm-token	FCM token registration and update
GET	children/me	Fetch child profile data
POST	locations/telemetry/{childId}	Upload location data
POST	apps/add-bulk	Report installed applications
POST	sessions	Upload session summary (captures, blocked/safe c
POST	sessions/{sessionId}/results	Upload individual analysis results (decisions, threat
GET	sessions/{sessionId}/results	Fetch analysis results for parent preview
GET	sessions	Fetch session history for behavior reports
GET	policies	Fetch active restriction policies
POST	rewards/claim	Claim earned reward
GET	rewards/balance	Fetch current reward state
POST	tasks/complete	Mark task as completed
GET	tasks	Fetch assigned tasks

8.4 Offline Operation

The application is designed to function reliably without persistent network connectivity:

- Local policy enforcement: Restriction policies are cached locally and enforced even when the backend is unreachable. Policy validity is time-stamped so stale policies can be flagged.
- Offline data queue: Location telemetry, installed apps data, and analysis results are stored locally with an isSent flag. A WorkManager task processes the queue when connectivity is restored.

- Exponential backoff: Failed API operations retry with exponential backoff (base 30 seconds, max 60 seconds, 5 retries maximum) to avoid hammering the backend during connectivity issues.
- Synchronization recovery: On connectivity restore (detected via ConnectivityReceiver), all pending operations are processed in priority order: policy sync first (to ensure correct enforcement), then telemetry and app data, then rewards.
- Cached authentication: The access token is stored securely in EncryptedSharedPreferences, allowing the application to authenticate immediately when connectivity is restored without re-pairing.

9. Dependency Injection with Hilt

9.1 DI Architecture

The application uses Hilt (2.51.1) for compile-time dependency injection. Hilt provides a standardized way to manage dependencies across the application's component hierarchy:

- Application Component (@HiltAndroidApp): Singleton-scoped, created once for the application lifetime. Hosts core dependencies that must be shared across all components.
- Activity Component (@AndroidEntryPoint): Activity-scoped, created and destroyed with MainActivity. Provides activity-level dependencies and ViewModels.
- Service Component: Service-scoped, created and destroyed with foreground services. Provides service-level dependencies.
- ViewModelComponent: Automatically scoped to each ViewModel's lifecycle via @HiltViewModel.

9.2 Provided Dependencies

Dependency	Scope	Module
Room Database	Singleton	AppModule
SessionDao	Singleton	AppModule
AnalysisResultDao	Singleton	AppModule
LocationTelemetryDao	Singleton	AppModule
Retrofit + OkHttpClient	Singleton	AppModule
ApiService	Singleton	AppModule
EncryptedSharedPreferences	Singleton	AppModule
AiAnalyzer (ONNX Runtime)	Singleton	AiModule
SessionManager	Singleton	AiModule
PreferenceManager	Singleton	AppModule
LogManager	Singleton	AppModule
AuthRepository	Singleton	AppModule
LocationRepository	Singleton	AppModule
FCMRepository	Singleton	AppModule
AppsRepository	Singleton	AppModule
RewardRepository	Singleton	AppModule
PolicyRepository	Singleton	AppModule

9.3 Module Organization

AppModule (@Module @InstallIn(SingletonComponent::class)):

Provides application-wide singletons:

- Room database instance and all DAOs.

- OkHttpClient with configured interceptors.
- Retrofit instance with kotlinx.serialization converter.
- EncryptedSharedPreferences instance.
- PreferenceManager and LogManager.
- All repository implementations.

AiModule (@Module @InstallIn(SingletonComponent::class)):

Provides AI-related dependencies that are computationally expensive to create:

- AiAnalyzer: Loads the ONNX model and embeddings from assets. Created lazily with @LazySingleton to defer model loading until first analysis request.
- SessionManager: Coordinates session lifecycle. Singleton to maintain session state across service restarts.

ServiceModule (@Module @InstallIn(ServiceComponent::class)):

Provides service-scoped dependencies for foreground services:

- MediaProjection instance.
- VirtualDisplay configuration.
- Capture loop dependencies.

ViewModelModule (@Module @InstallIn(ViewModelComponent::class)):

Provides ViewModel-scoped dependencies:

- ViewModel instances via @HiltViewModel annotation.
- Automatic disposal of coroutine scopes on ViewModel clear.

9.4 Benefits of Hilt in This Application

- Lifecycle-safe service injection: Foreground services receive their dependencies through Hilt's service component, ensuring proper cleanup when the service is destroyed.
- Singleton AI model: The AiAnalyzer loads the ONNX model once and shares it across all sessions, avoiding repeated 100+ MB model loads.
- Testability: Dependencies can be replaced with mocks in tests through Hilt testing support.
- Compile-time safety: Dependency graph validation happens at compile time, preventing runtime injection failures.
- Consistent scoping: Repository instances are singletons, ensuring consistent data access across ViewModels and services.

10. Security and Reliability

10.1 Authentication

The application uses JWT (JSON Web Token)-based authentication for all backend communication:

- Token acquisition: A token is obtained during the device pairing process (POST children/pair) and stored in EncryptedSharedPreferences.
- Token storage: The access token is encrypted using AES-256-GCM via the Android Security Crypto library before being written to SharedPreferences. The encryption key is stored in the Android KeyStore, which provides hardware-backed security on supported devices.
- Token injection: Every API request includes the token via the AuthInterceptor, which reads the token from EncryptedSharedPreferences and adds an Authorization: Bearer <token> header.
- Token lifecycle: Tokens are validated on each API call. If the backend returns a 401 Unauthorized status, the SafeApiCall wrapper returns an error state, and the application can trigger re-authentication if needed.

10.2 Secure Communication

- TLS encryption: All API communication occurs over HTTPS with TLS 1.3, ensuring data-in-transit protection.
- Certificate pinning (optional): OkHttp can be configured with CertificatePinner to restrict trusted certificates

to specific pins, providing additional protection against man-in-the-middle attacks.

- API request validation: The backend validates all requests for structural integrity and authorization before processing.

10.3 Secure Local Storage

Data	Storage Mechanism	Encryption
Access token	EncryptedSharedPreferences	AES-256-GCM
FCM token	EncryptedSharedPreferences	AES-256-GCM
Child profile	EncryptedSharedPreferences	AES-256-GCM
PIN hash	EncryptedSharedPreferences (SHA-256 + sa	Not reversibly encrypted
User preferences	DataStore	Not encrypted (non-sensitive)
Location data	Room database	Database-level (optional)
Analysis results	Room database	Database-level (optional)
Captured images	Internal storage	App-specific directory (not accessible to other apps

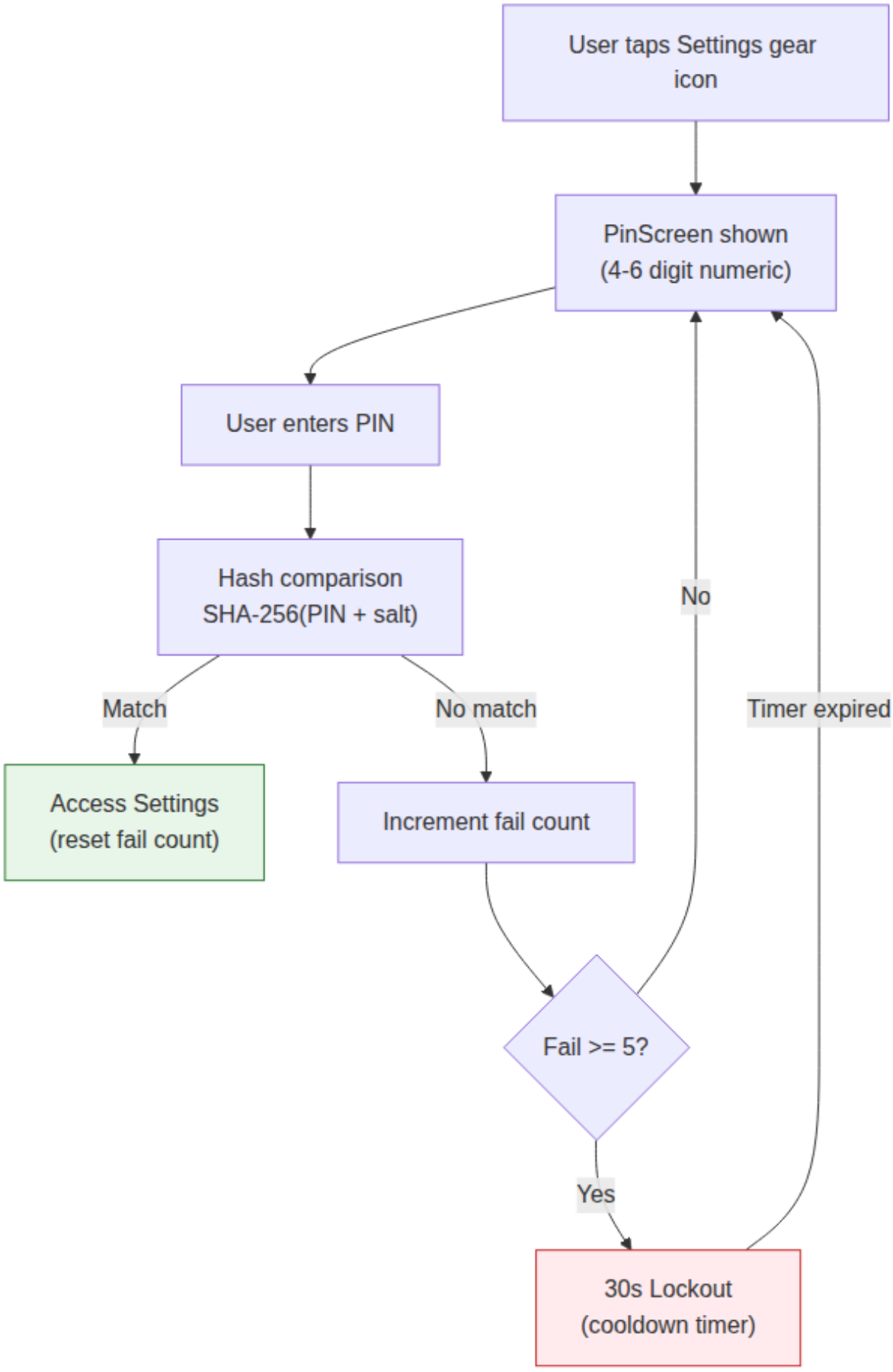
10.4 Offline PIN Protection

To prevent unauthorized access to application settings when the device is offline and parent authentication via the backend is unavailable, a local PIN system gates access to the Settings screen.

PIN Creation and Storage:

- The PIN is set during initial device setup after QR pairing.
- The PIN hash (SHA-256 with a random salt) is stored in EncryptedSharedPreferences.
- The salt is generated using SecureRandom and stored alongside the hash.
- The raw PIN is never stored. Only the hash is persisted.

PIN Entry Flow:



Attempt Tracking:

- Failed attempts are tracked in EncryptedSharedPreferences (failedAttempts counter).
- After 5 consecutive failures, a 30-second cooldown is enforced (lockoutUntil timestamp).
- On successful PIN entry, the failedAttempts counter is reset to zero.
- The lockout time is validated on the client side; the Settings screen remains inaccessible until the cooldown expires.

PIN Recovery:

If the parent forgets the PIN, recovery requires re-pairing:

- The parent generates a new QR code through the parent application.
- The child scans the QR code, which includes an optional PIN reset flag.
- The PIN hash in EncryptedSharedPreferences is cleared.
- A new PIN is set during the re-pairing flow.

Screens Not Protected by PIN:

- Task Screen: Accessible freely so the child can track responsibilities.
- Reward Screen: Accessible freely so the child can view progress and motivation.

Screens Protected by PIN:

- Settings Screen: Changing monitoring preferences, viewing logs, adjusting permissions, toggling monitoring state.

Implementation:

- PinManager: Singleton managing PIN hash generation, verification, and attempt tracking.
- PinScreen: Compose screen with numeric keypad, visual feedback, error state, and lockout timer.
- PinViewModel: Manages PIN entry state, validation logic, and lockout state with StateFlow.
- EncryptedSharedPreferences: Stores pinHash, pinSalt, failedAttempts, lockoutUntil.

10.5 Tamper Resistance

- Service auto-restart : Foreground services restart automatically if killed by the system, with up to 5 consecutive restart attempts and exponential backoff.
- Boot persistence: The BootReceiver restarts monitoring and location services when the device reboots, ensuring continuous protection.
- Permission verification: On every service start, all required permissions are re-verified. If permissions have been revoked, the service enters a PermissionRequired state and notifies the user.
- Integrity checks: The application performs periodic validation of its core components. If the Accessibility Service is disabled or the monitoring service is not running, the user is notified and prompted to re-enable.

10.6 Reliability Measures

Measure	Implementation
Crash recovery	Automatic service restart with exponential backoff (max 5 attempts)
Network retry	Exponential backoff (30s base, 60s max, 5 retries)
Database performance	Batched writes for SAFE results (10 items or 5 seconds)
Capture fault tolerance	Consecutive failure threshold with max retry cap, service auto-restart
Background persistence	High-priority foreground notification keeps process alive
Offline data queue	Unsent data persisted locally with automatic sync on connectivity restore
ANR prevention	All network and AI operations on background coroutines

10.7 Error Handling and Logging

API Error Handling

All API calls are wrapped in a SafeApiCall inline function that returns a sealed ApiResult type:

```
sealed class ApiResult<out T> {
    data class Success<T>(val data: T) : ApiResult<T>()
    data class Error(
        val message: String,
        val code: Int? = null,
        val details: String? = null
    ) : ApiResult<Nothing>()
}
```

The wrapper catches three exception types:

- HttpException: Server errors (4xx/5xx) with status code extraction.
- IOException: Network failures (timeout, DNS, connectivity loss).
- Generic Exception: Unexpected errors with message capture.

HTTP Logging Interceptor

A custom AppLoggingInterceptor logs every API request and response including HTTP method, URL path, request body, response status code, elapsed time in milliseconds, and response body. All logs are routed through LogManager for in-app viewing.

Retry and Backoff

Failed API operations implement exponential backoff with base delay of 30 seconds, maximum delay of 60 seconds, and maximum of 5 retries. This applies to location telemetry upload, FCM token registration, and apps synchronization.

In-App Logging (LogManager)

LogManager provides 5 log types stored in SharedPreferences with a capacity of up to 100 entries:

Log Type	Color	Use Case
INFO	White	General events
SUCCESS	Green	Successful API calls and operations
ERROR	Red	Failures and exceptions
LOCATION	Blue	GPS acquisition and telemetry
HTTP	Yellow	Request and response details

Logs are viewable in the Settings screen's collapsible LogSection with monospace formatting and color-coded rows.

Crash Recovery

- Foreground services auto-restart on crash with up to 5 consecutive attempts.
- Exponential backoff between restart attempts.
- BootReceiver restarts services after device reboot.
- Permission re-verification on every service start.

10.8 Anti-Uninstall and Anti-Disable

The application employs multiple layers to prevent the child from uninstalling or disabling the app, ensuring continuous protection cannot be bypassed.

Device Admin Enrollment

The app registers as a Device Admin via DeviceAdminReceiver:

```
class SecurityAdminReceiver : DeviceAdminReceiver() {
    override fun onEnabled(context: Context, intent: Intent) {
        LogManager.log("Device admin granted", LogType.INFO)
    }

    override fun onDisabled(context: Context, intent: Intent) {
```

```
            LogManager.log("Device admin revoked — alerting parent", LogType.ERROR)
            FcmNotificationSender.sendAdminDisabledAlert(context)
        }
    }
```

Enrollment is requested after QR pairing completes. While Device Admin is active:

- The standard Android uninstall flow is blocked — the user must first deactivate the admin in Security settings.
- Deactivating admin triggers onDisabled, which sends an immediate FCM alert to the parent.
- The parent is notified that the child has attempted to disable protection.

Accessibility Service Uninstall Interception

The Accessibility Service monitors for navigation to package management screens:

- Detects when the system Settings App Info page is opened for this application's package name.
- Intercepts the TYPE_WINDOW_STATE_CHANGED event for package installer and settings packages.
- Displays a PIN-gated blocking overlay preventing access to the uninstall or force-stop button.
- If Accessibility Service is revoked, an FCM notification is sent to the parent.

Screen Pinning

For younger children, the app supports optional screen pinning:

- The parent enables pinning through the parent application.
- The child device calls startLockTask() to pin the current task.
- Exiting the pinned state requires a PIN (set during pairing).
- If unpinning is attempted without the PIN, a notification is sent to the parent.

Integrity Monitoring

A periodic WorkManager task monitors critical components:

Component	Check	Action on Failure
Foreground Service	`isServiceRunning()`	Auto-restart with exponential backoff
Accessibility Service	`isAccessibilityEnabled()`	Request re-enable via system prompt
Device Admin	`isAdminActive()`	Notify parent via FCM
Screen Pinning	`isLockTaskActive()` (if enabled)	Re-pin with PIN challenge

Checks run every 15 minutes. On repeated failures (3+ consecutive), an immediate FCM alert is escalated to the parent.

Testing and Evaluation

1. Testing Strategy

The testing strategy follows the Android testing pyramid with coverage across unit, integration, instrumentation, and end-to-end levels:

Level	Scope	Tools
Unit Testing	ViewModels, Repositories, Utility classes	JUnit 5, MockK, Turbine
Integration Testing	Repository + Database, API + Network	Room testing, MockWebServer
AI Pipeline Testing	ONNX inference accuracy, OCR accuracy	Custom test dataset, AssertJ
Instrumentation Testing	Foreground services, Workers, Permissions	AndroidX Test, WorkManagerTestHelper
End-to-End Testing	Full user flows, offline resilience	adb, custom scripts

2. Unit Testing

ViewModel Tests

Each ViewModel is tested for state transitions, error handling, and edge cases:

- PairingViewModel: QR code parsing with valid and invalid JSON, successful and failed pairing requests, UI state transitions (Scanning -> Loading -> Success/Error).
- HomeViewModel: Location sending with success and network failure, installed apps sync, child profile fetch with cached fallback.
- PinViewModel: Valid PIN acceptance, invalid PIN rejection, failed attempt increment, lockout after 5 failures, lockout expiry.
- SettingsViewModel: Preference toggle persistence with DataStore, permission state mapping for all runtime permission states (granted, denied, permanently denied).

Repository Tests

Repositories are tested with mocked API service and in-memory Room database:

- AuthRepository: Successful pairing, network failure with cached fallback, invalid token rejection.
- LocationRepository: Successful telemetry upload, offline queue fallback with isSent flag verification, retry logic after failure.
- AppsRepository: PackageManager query with mocked packages, bulk upload with partial failure handling.
- FCMRepository: Initial token registration, re-registration on token change, server error handling.

Utility Tests

- SafeApiCall: Success path returning ApiResult.Success, HttpException mapping with status codes, IOException mapping, generic exception catching.
- LogManager: Log insertion capacity (100 entries), overflow eviction (FIFO), clear operation, formatTimestamp correctness.
- PreferenceManager: Read/write for each stored field type (String, Boolean, Int, Long), default value fallback, missing key handling.
- ApiConfig: URL construction correctness, endpoint path derivation.

3. AI Pipeline Validation

OCR Accuracy Testing

- Test set: Images containing banned words, safe text, and mixed content across all 7 banned categories.
- Metrics: Text detection rate, banned word recall, false positive rate.
- Early exit validation: Verification that banned word detection correctly skips ONNX inference, with timing measurements to confirm the performance benefit.
- Test set: Images containing banned words, safe text, and mixed content across all 7 banned categories.
- Metrics: Text detection rate, banned word recall, false positive rate.
- Early exit validation: Verification that banned word detection correctly skips ONNX inference, with timing measurements to confirm the performance benefit.

Performance Benchmarks

Continuous integration benchmarks enforced on each pull request:

Metric	Target	Measurement Method
OCR latency	< 250 ms	ocrTimeMs instrumentation
ONNX inference latency	< 500 ms	onnxTimeMs instrumentation
Total analysis latency	< 1000 ms	Pipeline timing instrumentation
Memory during session	< 250 MB PSS	dumpsys meminfo
Frame capture rate	>= 95% of configured interval	Capture success / total attempts
Blur FSM response time	< 100 ms	FSM transition instrumentation

4. Integration Testing

Database Integration

- Room DAO tests: Insert, read (by ID, by session, all), update, and delete operations for each entity using in-memory Room database.
- Migration tests: Schema migration from version 1 to version 2 verified with data preservation assertions.
- Batched write tests: Verification that batched writes correctly flush after 10 results or 5 seconds, and that mixed SAFE/BLOCKED writes do not interleave incorrectly.

Network Integration

- MockWebServer: Simulate API endpoints with success responses, error responses (4xx, 5xx), and timeout scenarios.
- AuthInterceptor test: Verify Bearer token injection for authenticated requests and missing-token handling for unauthenticated requests.
- AppLoggingInterceptor test: Verify that request and response data is correctly logged to LogManager with proper log type attribution.

5. Instrumentation Testing

Service Tests

- Foreground service lifecycle: Start service, verify active state, stop service, verify completed state, restart after simulated crash.
- Capture loop: Frame acquisition rate at configured intervals, bitmap extraction correctness, error recovery when ImageReader returns null.
- Blur FSM: Full state transition verification for SAFE -> PENDING_BLUR -> BLURRED -> PENDING_RELEASE -> SAFE cycle with configurable thresholds.
- Session persistence: Verify that session data survives service restart and process kill.

Permission Tests

- Camera permission: Grant, deny, and permanently deny flows for QR pairing screen.
- Location permission: Fine location, coarse location, and background location permission states.
- MediaProjection: Intent result handling when user accepts or declines screen capture consent dialog.
- Overlay permission: Settings intent navigation, grant detection, and re-check flows.

6. End-to-End Testing

- Pairing flow: Launch application -> QR scanning screen -> scan valid QR -> API call -> token storage -> navigate to Home Screen.
- Monitoring session: Start monitoring -> capture frames -> run AI analysis -> store results -> stop monitoring -> verify session data integrity.
- Offline resilience: Enable airplane mode -> capture continues in offline mode -> data queued in Room -> reconnect network -> verify data synchronizes.
- FCM trigger: Send silent push notification from test server -> application receives in FCMService -> verify immediate location sync triggered.
- PIN gate flow: Navigate to Settings -> PIN screen displayed -> enter wrong PIN -> increment counter -> enter correct PIN -> access Settings.

7. Testing Tools and Dependencies

Tool	Purpose
JUnit 5	Test runner and assertion framework
MockK	Kotlin-specific mocking library
Turbine	Flow and StateFlow testing
MockWebServer (OkHttp)	HTTP endpoint simulation
Room Testing	In-memory database for DAO tests
Compose UI Test	Jetpack Compose screen interaction testing
AndroidX Test	Instrumentation test orchestration
WorkManagerTestHelper	Worker unit testing
adb + dumpsys	Performance benchmarking
Custom timing instrumentation	AI pipeline per-phase measurement

8. Android Performance Benchmarks

Full Pipeline Latency (Android, 856 frames)

Operation	Average	Min	Max	Std Dev
Capture (ImageReader)	0.31 ms	0.14 ms	2.37 ms	0.28 ms
Bitmap Conversion	6.55 ms	3.63 ms	50.18 ms	3.12 ms
Analysis (ONNX + OCR)	609.98 ms	195.35 ms	1631.46 ms	312.45 ms

OCR early exit: When banned words are detected, analysis exits early reducing processing time to ~5-50 ms.

Memory Consumption

Metric	Value
Average PSS Memory	184.6 MB

Peak PSS Memory	~200 MB

Battery Consumption

Metric	Value
Test Duration	15m 35s
Battery Change	100% to 96% (-4%)
Total Discharge	197 mAh
App Power Contribution	31.6 mAh

Test Environment

Parameter	Specification
Device	Google Pixel 6
Android Version	Android 15 (API 35)
Memory	8 GB LPDDR5
Processor	Google Tensor (5 nm)
Test Duration per Session	15.5 minutes
Frames per Session	~900
Session Interval	1000 ms
Network	Wi-Fi (stable)

Edge Case Validation

- Empty capture frames: When MediaProjection returns a null image, the system retries up to 3 times before marking the frame as an error capture.
- Permission revocation during session: If the user revokes MediaProjection or SYSTEM_ALERT_WINDOW permission mid-session, the FSM transitions to PermissionRequired state and pauses capture.
- Rapid app switching: The Accessibility Service correctly tracks foreground app changes even under rapid switching conditions (tested at 10 switches per second).
- Concurrent notification storms: The notification system handles up to 50 simultaneous blocked content alerts without ANR or crash.
- Database corruption recovery: If the Room database file is corrupted, the application falls back to a clean state with data loss limited to the current session.